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Fundamentals of statistical learning

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1 Defining probability

1.1 Why probability?

Observed data are the results of processes whose behavior we may not fully know. To reason about them, we describe the process with a probabilistic model. Probability lets us work forward from a model to possible outcomes; statistical inference uses observed outcomes to learn about the model.

Example 1: A coin toss

Let $\theta = P(\text{heads})$ be the probability that a coin lands heads. If θ is known, we can ask how likely it is to observe three heads in five tosses. If instead we observe three heads in five tosses while θ is unknown, we can use those data to learn about θ.

1.2 Sample spaces and events

Before assigning probabilities, we must specify what the experiment can produce. We start with a finite set of possible results.

Definition 1.1: Outcomes and events

A random experiment is a process whose result is not known in advance. Each possible result is an *outcome* ω. The set of all possible outcomes is the *sample space* Ω. An *event* is a subset $A \subseteq \Omega$; it occurs when the observed outcome belongs to A.

Example 2: Rolling a die

For one roll of a six-sided die, the sample space is $\Omega = \{1, 2, 3, 4, 5, 6\}$. The event "the result is even" is $A = \{2, 4, 6\}$. The whole sample space Ω is the certain event, while $\varnothing$ is the impossible event.

1.2.1 Operations on events

Events are subsets of the same sample space Ω, so set operations let us describe new events in terms of A and B:

- **Union:** $A \cup B := \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B\}$. This event occurs when at least one of A and B occurs; "or" includes the case in which both occur. In particular, $A \cup \varnothing = A$ and $A \cup \Omega = \Omega$.
- **Intersection:** $A \cap B := \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\}$. This event occurs when both A and B occur. In particular, $A \cap \Omega = A$ and $A \cap \varnothing = \varnothing$.
- **Complement:** $A^c := \Omega \setminus A = \{\omega \in \Omega : \omega \notin A\}$. This event occurs when A does not occur. In particular, $A \cup A^c = \Omega$ and $A \cap A^c = \varnothing$.

For the die in Example 2, let $B = \{4, 5, 6\}$ be the event "the result is at least four". Then $A \cup B = \{2, 4, 5, 6\}$, $A \cap B = \{4, 6\}$, and $A^c = \{1, 3, 5\}$.

Two events are *disjoint* when $A \cap B = \varnothing$, so they cannot occur together. If $A \subseteq B$, every outcome in A is also in B: whenever A occurs, B occurs.

Proposition 1.1: De Morgan's laws

For any events A and B,

$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$

More generally, for events $A_1, \ldots, A_n$,

$$\left(\bigcup_{i=1}^{n} A_i\right)^c = \bigcap_{i=1}^{n} A_i^c, \qquad \left(\bigcap_{i=1}^{n} A_i\right)^c = \bigcup_{i=1}^{n} A_i^c.$$

The first identity says that "none occurs" means that every event fails. The second says that "not all occur" means that at least one event fails. We will use the second identity in the proof of Bonferroni's inequality.

1.2.2 Partitions and event decomposition

A useful way to study an event is to split it into disjoint pieces. For any events A and B, exactly one of B and B^c occurs, so

$$A = A \cap \Omega = A \cap (B \cup B^c) = (A \cap B) \cup (A \cap B^c).$$

The two events on the right are disjoint. For example, a car can fail an emissions test either while it pollutes or while it does not.

More generally, a *partition* $B_1, \ldots, B_n$ of Ω consists of pairwise disjoint events whose union is Ω. Every event A can then be written as

$$A = \bigcup_{i=1}^{n} (A \cap B_i), \qquad (A \cap B_i) \cap (A \cap B_j) = \varnothing \quad (i \neq j).$$

This decomposition will let us add the probabilities of different ways in which the same event can occur.

1.3 Probability

There are several ways to interpret a probability. Under a *symmetry* assumption, equally likely outcomes give $P(A) = |A|/|\Omega|$. A *frequency* interpretation concerns the long-run proportion of repetitions in which A occurs. A *subjective* interpretation expresses a degree of belief about A. The axioms below state the mathematical rules that a probability assignment must satisfy, regardless of its interpretation.

Example 3: Two dice

If the 36 ordered results of rolling two dice are equally likely, they form the sample space $\Omega = \{1, \dots, 6\}^2$. The sum is eight for $(2,6), (3,5), (4,4), (5,3), (6,2)$, so the probability of that event is $5/36$. Counting outcomes this way is valid because the ordered pairs are equally likely.

Definition 1.2: Probability on a finite sample space

A probability measure P assigns a number $P(A)$ to every event $A \subseteq \Omega$ such that:

1. $0 \le P(A) \le 1$ for every event A;
2. $P(\Omega) = 1$;
3. if $A \cap B = \varnothing$, then $P(A \cup B) = P(A) + P(B)$.

The last property is called *finite additivity*.

Because A and A^c are disjoint and their union is Ω, these axioms give $P(A^c) = 1 - P(A)$. In particular, $P(\varnothing) = 0$.

They also imply *monotonicity*: if $A \subseteq B$, then $B = A \cup (B \cap A^c)$ is a disjoint union, so $P(B) = P(A) + P(B \cap A^c) \ge P(A)$.

Example 4: A fair die

If the six outcomes of a die roll are equally likely, each has probability $1/6$. The event $A = \{2, 4, 6\}$ therefore has probability $P(A) = 3/6 = 1/2$.

For an infinite sample space, finite additivity is replaced by *countable additivity*: for pairwise disjoint events $A_1, A_2, \dots$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

In that setting, a probability measure is defined on a suitable family of events rather than necessarily on every subset of Ω. On a finite sample space, the finite formulation above is enough.

1.3.1 Inclusion–exclusion

Finite additivity directly handles disjoint events. If A and B overlap, adding $P(A)$ and $P(B)$ counts their intersection twice.

Proposition 1.2: Inclusion–exclusion for two events

For any events A and B,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Proof. Split $A \cup B$ into the disjoint events A and $B \cap A^c$:

$$P(A \cup B) = P(A) + P(B \cap A^c).$$

The event B is itself the disjoint union of $A \cap B$ and $A^c \cap B$. Thus $P(B \cap A^c) = P(B) - P(A \cap B)$, which gives the formula. □

Example 5: Overlapping die events

Let $A = \{2, 4, 6\}$ and $B = \{4, 5, 6\}$. Then $P(A) = P(B) = 1/2$ and $P(A \cap B) = 1/3$. Therefore

$$P(A \cup B) = \frac{1}{2} + \frac{1}{2} - \frac{1}{3} = \frac{2}{3},$$

in agreement with $A \cup B = \{2, 4, 5, 6\}$.

Applying the two-event formula twice gives the three-event version:

$$\begin{aligned} P(A \cup B \cup C) = {} & P(A) + P(B) + P(C) \\ & - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ & + P(A \cap B \cap C). \end{aligned}$$

The final term restores outcomes in all three events: they were added three times and subtracted three times.

For n events, the same pattern continues:

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{\varnothing \neq I \subseteq \{1,\dots,n\}} (-1)^{|I|+1} P\left(\bigcap_{i \in I} A_i\right).$$

To see why, consider an outcome contained in exactly k of the events. It is counted $\binom{k}{j}$ times among the intersections of size j, so its total coefficient on the right is $\sum_{j=1}^{k} (-1)^{j+1} \binom{k}{j} = 1$. An outcome in none of the events contributes zero to both sides.

1.3.2 Probability bounds

Sometimes the probabilities of intersections are unknown. We can still bound the probability of a union.

Proposition 1.3: Union bound (Boole's inequality)

For events $A_1, \ldots, A_n$,

$$P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i).$$

Proof. We prove the bound by induction on the number of events.

Base case ($n = 1$). The union contains only A_1, so the inequality is an equality: $P(A_1) = \sum_{i=1}^{1} P(A_i)$.

Inductive step. Assume the bound holds for n events and set $U_n = \bigcup_{i=1}^{n} A_i$. The union of $n+1$ events is $U_n \cup A_{n+1}$. Applying inclusion–exclusion gives

$$\begin{aligned} P\left(\bigcup_{i=1}^{n+1} A_i\right) &= P(U_n) + P(A_{n+1}) - P(U_n \cap A_{n+1}) \\ &\leq P(U_n) + P(A_{n+1}) \\ &\leq \sum_{i=1}^{n} P(A_i) + P(A_{n+1}) = \sum_{i=1}^{n+1} P(A_i). \end{aligned}$$

The first inequality uses $P(U_n \cap A_{n+1}) \geq 0$; the second uses the induction hypothesis. Thus the bound holds for every $n \geq 1$. □

For a countable sequence, the same inequality follows by applying the finite bound to the first n events and taking the limit.

Proposition 1.4: Bonferroni's inequality

For events $A_1, \ldots, A_n$,

$$P\left(\bigcap_{i=1}^{n} A_i\right) \geq \sum_{i=1}^{n} P(A_i) - (n-1).$$

Proof. The second n-event form of De Morgan's law turns the complement of an intersection into a union of complements. Then the union bound gives

$$\begin{aligned} P\left(\bigcap_{i=1}^{n} A_i\right) &= 1 - P\left(\bigcup_{i=1}^{n} A_i^c\right) \\ &\geq 1 - \sum_{i=1}^{n} P(A_i^c) \\ &= 1 - \sum_{i=1}^{n} (1 - P(A_i)) = \sum_{i=1}^{n} P(A_i) - (n-1). \end{aligned}$$

□

1.3.3 Application: a randomized identity test

Suppose F and G are polynomials of degree at most d. We want to test whether they are identical without comparing all their coefficients. Choose a nonempty finite set S of possible evaluation points.

Algorithm 1.1: Randomized polynomial identity test

Given F, G, and S, report whether the polynomials differ or may be identical.

```
function IdentityTest(F, G, S)
    Draw R uniformly at random from S
    if F(R) ≠ G(R) then
        return DIFFERENT
    else
        return POSSIBLY EQUAL
    end if
end function
```

If $F = G$, the test never reports a difference. If $F \neq G$, it fails to detect the difference only when R belongs to the event

$$E = \{r \in S : F(r) = G(r)\}.$$

Indeed, E contains the roots in S of the nonzero polynomial $H = F - G$, which has at most d roots. Because R is uniform on S,

$$P(R \in E) = \frac{|E|}{|S|} \leq \frac{d}{|S|}.$$

For $S = \{1, \ldots, 100d\}$ with $d \geq 1$, the probability of missing a difference is at most $1/100$. The possible error is *one-sided*: "different" is certain, while "possibly equal" is not a proof of identity.

2

Conditional probability

2.1 Understanding conditional probability

Knowing that an event B occurred restricts our attention to its outcomes. Within this smaller group, how often does A occur?

Definition 2.1: Conditional probability

For events A, B with $P(B) > 0$, the probability of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

For a fair die, let A mean an even result and B a result greater than 3. Among the three outcomes in $B = \{4, 5, 6\}$, two are even: $P(A \mid B) = (2/6)/(3/6) = 2/3$. Rearranging the definition gives the **multiplication rule**:

$$P(A \cap B) = P(A \mid B)P(B).$$

Within the conditioning group, A and A^c remain complementary, so $P(A^c \mid B) = 1 - P(A \mid B)$.

2.1.1 Odds

For an event A with $P(A) < 1$, its *odds in favor* are

$$O(A) := \frac{P(A)}{P(A^c)} = \frac{P(A)}{1 - P(A)}.$$

Conversely, $P(A) = O(A)/(1 + O(A))$. For a fair die, the odds in favor of rolling a six are $(1/6)/(5/6) = 1/5$, often stated as one to five. This is different from saying that the probability is $1/5$; the probability is $1/6$.

2.2 Independent events

Definition 2.2: Independent events

Two events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

When $P(B) > 0$, this is equivalent to $P(A \mid B) = P(A)$: knowing that B occurred does not change the probability of A.

For more than two events, we need **mutual independence**: every subcollection must satisfy the product rule. For events $A_1, \ldots, A_n$, this means

$$P\left(\bigcap_{i \in J} A_i\right) = \prod_{i \in J} P(A_i) \quad \text{for every nonempty } J \subseteq \{1, \ldots, n\}.$$

The same product rule holds when selected events are replaced by their complements. In particular, independent coin tosses allow us to multiply the probabilities of the specified heads and tails.

2.2.1 Conditional independence

Events A and B are **conditionally independent given** C, with $P(C) > 0$, when

$$P(A \cap B \mid C) = P(A \mid C)P(B \mid C).$$

We apply independence within the group C. This does not assert that A and B are independent before conditioning.

2.3 Bayes' theorem

2.3.1 The law of total probability

Let $B_1, \ldots, B_m$ form a partition of the sample space, with $P(B_i) > 0$. The events $A \cap B_i$ are disjoint and together make up A. Addition of their probabilities and the multiplication rule therefore give

$$P(A) = \sum_{i=1}^{m} P(A \cap B_i) = \sum_{i=1}^{m} P(A \mid B_i)P(B_i).$$

Groups of probability zero can be omitted. This formula computes an overall probability by combining the probabilities within the separate groups.

2.3.2 Reversing the conditioning

Theorem 2.1: Bayes' theorem

For a group B_j of the partition and an event A with $P(A) > 0$,

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{P(A)} = \frac{P(A \mid B_j)P(B_j)}{\sum_{i=1}^{m} P(A \mid B_i)P(B_i)}.$$

Proof. The conditional probability definition gives $P(B_j \mid A) = P(B_j \cap A)/P(A)$. The multiplication rule replaces its numerator by $P(A \mid B_j)P(B_j)$, and total probability supplies the denominator. □

We start with the probability $P(B_j)$ before observing A, called the **prior**, and update it to $P(B_j \mid A)$, called the **posterior**. For example, a disease is a group and a positive test is the observed event; their probabilities are worked out in the exercise appendix.

3

Random variables

3.1 From outcomes to data

Imagine picking one student at random from a class. Before the draw, we do not know who we will get; afterwards, we can write down that student's height, weight, and age. Probability describes the draw, while the numbers we write down are the data.

The possible outcomes form Ω, and $\omega \in \Omega$ is the outcome that actually occurs. We write $(\Omega, \mathcal{F}, P)$ for the probability space: $\mathcal{F}$ contains the events to which P assigns probabilities. For a finite Ω, as in Chapter 1, every subset can be an event.

Definition (informal): Random object

Given a probability space, a *random object* is a map

$$Z : \Omega \longrightarrow \mathcal{T},$$

where $\mathcal{T}$ is a suitable space of observations for our data-collection process.

Definition 3.1: Random object

A *random object* is a measurable* map from outcomes to possible observations:

$$Z : \Omega \longrightarrow \mathcal{T}, \qquad \omega \longmapsto Z(\omega).$$

The set $\mathcal{T}$ says what kind of thing we record. It could contain numbers, vectors, images, or text messages. Once ω is known, $Z(\omega)$ is the observation we get.

*A rigorous definition of measurability is beyond the scope of this course.

Think of Z as the rule that turns the outcome into a piece of data. "Measurable" is the technical condition that lets us assign a probability to questions about that data, such as "is the selected student taller than 180 cm?"

3.2 Random variables and random vectors

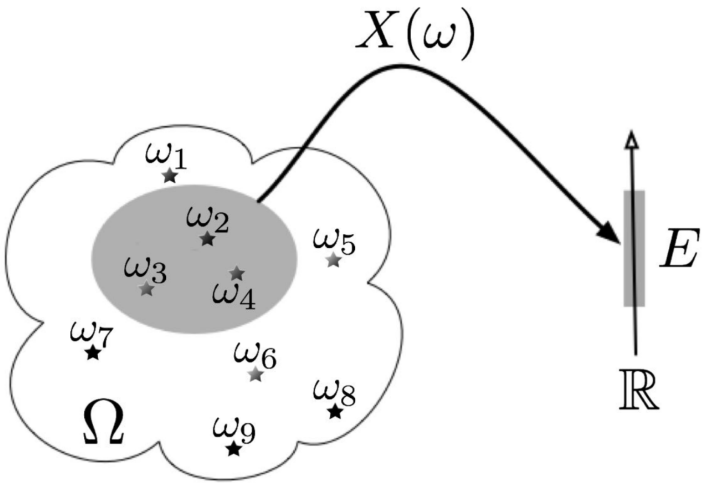
Often the observation is a number or a list of numbers. These are two particularly useful kinds of random objects.

Definition 3.2: Random variable

A *random variable* is a measurable map

$$X : \Omega \longrightarrow \mathbb{R}.$$

It assigns one real number $X(\omega)$ to each outcome ω.



A map from Ω to $\mathbb{R}$. Illustration from Brutti's slides.

Definition 3.3: Random vector

A *p-dimensional random vector* is a measurable map

$$\underline{X} : \Omega \longrightarrow \mathbb{R}^p, \qquad \underline{X}(\omega) := \big(X_1(\omega), \ldots, X_p(\omega)\big).$$

Each coordinate X_j is a real-valued random variable. Thus one outcome gives us p numbers collected into a vector. These measurements can be related; they do not have to be independent.

Note: Notation

Following Brutti's notation, we denote a random vector by $\underline{X}$ (X with a bar underneath):

$$\underline{X} = (X_1, \ldots, X_p).$$

We write $\underline{x} = (x_1, \ldots, x_p)$ for a vector of observed values. Individual components remain X_j and x_j, without a bar.

Example 6: One student, one number or three

Pick a student at random and let ω identify the student selected. If H records height in centimetres, then $H(\omega)$ is one number. For a student who is 178 cm tall, $H(\omega) = 178$.

Now record height H, weight W in kilograms, and age A in years for that *same* student:

$$\underline{X}(\omega) = \begin{pmatrix} H(\omega) \\ W(\omega) \\ A(\omega) \end{pmatrix} = \begin{pmatrix} 178 \\ 72 \\ 21 \end{pmatrix} \in \mathbb{R}^3.$$

The draw gives us one student, not three. We simply take three measurements of that student. Before the draw, H and $\underline{X}$ are uncertain; for the selected student, their values are fixed.

In supervised learning, each sampled item gives us an input $\underline{X}$ (the features) and a target Y (the label or quantity of interest). Before we see the item, both are uncertain; for an observed item, their values are fixed.

Example 7: House prices

Choose a house at random. Use its floor area in square metres, number of rooms, and age in years as features. For a particular house with 85 m^2, 3 rooms, and an age of 10 years, the data look like this:

$$\underline{X}(\omega) = (85, 3, 10), \qquad Y(\omega) = 240{,}000 \text{ euros}.$$

The three features make a random vector; the sale price is one random variable.

Example 8: Image labels

Take a randomly chosen 64×64 grayscale photo. Each pixel is one number from 0 (black) to 255 (white). Putting the pixels in a fixed order gives $\underline{X}(\omega) \in \mathbb{R}^{4096}$: one vector with 4096 entries for one photo. If we set $Y(\omega) = 1$ for a cat and $Y(\omega) = 0$ for a dog, then a cat photo gives one feature vector and the single label 1. The pixels can be related even though they are separate coordinates of $\underline{X}$.

3.3 Preimages and induced probabilities

To ask a probability question about the values of X, we first find the outcomes that produce those values.

Definition 3.4: Preimage

For a random variable $X : \Omega \to \mathbb{R}$ and a set $E \subseteq \mathbb{R}$, the *preimage* (or *inverse image*) of E is

$$X^{-1}(E) := \{\omega \in \Omega : X(\omega) \in E\} \subseteq \Omega.$$

To find this event, we collect the outcomes whose outputs fall in E. Whenever $X^{-1}(E)$ is an event, we define

$$\begin{aligned} P(X \in E) &:= P\big(X^{-1}(E)\big) \\ &= P\big(\{\omega \in \Omega : X(\omega) \in E\}\big). \end{aligned}$$

In particular, for a single value x,

$$P(X = x) := P\big(X^{-1}(\{x\})\big).$$

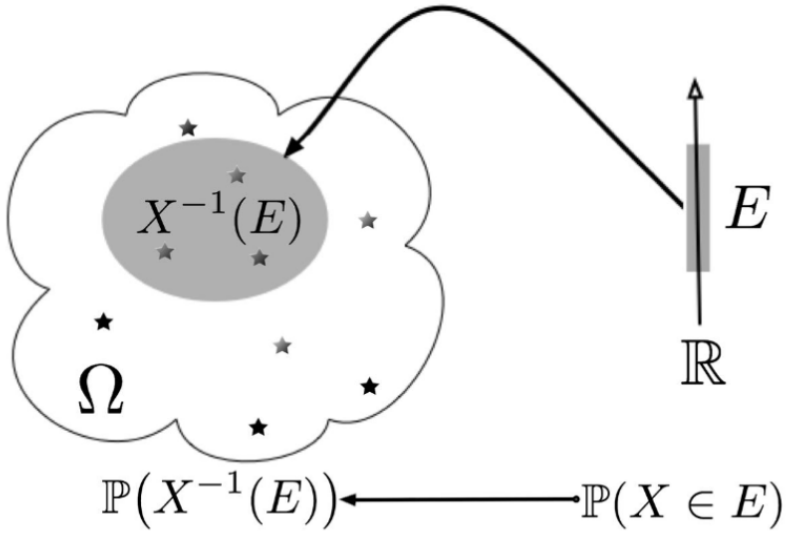
Here X is the random variable, while x is a particular real number.

Definition 3.5: Distribution of a random variable

The *distribution* of X is the rule that tells us with what probability X takes values in a given set E.

$$P_X(E) = P(X \in E).$$

The set E contains values; $X^{-1}(E)$ contains the outcomes that give those values. In our finite examples, every subset of Ω is an event.



A preimage and its probability. Illustration from Brutti's slides.

Example 9: Which students have heights in an interval?

Choose one student uniformly from $\Omega = \{\text{Anna}, \text{Luca}, \text{Sara}, \text{Marco}\}$. Let X record height in centimetres:

ω	Anna	Luca	Sara	Marco
$X(\omega)$	165	178	178	190

For $E = [170, 180]$, Luca and Sara are the students whose heights fall inside the interval. Thus

$$X^{-1}([170, 180]) = \{\text{Luca}, \text{Sara}\}, \qquad P(170 \le X \le 180) = \frac{2}{4} = \frac{1}{2}.$$

The same two students also give the exact value 178: $X^{-1}(\{178\}) = \{\text{Luca}, \text{Sara}\}$, so $P(X = 178) = 1/2$. One value can correspond to several outcomes.

Example 10: Counting heads in two coin tosses

Toss a fair coin twice, with independent tosses. The four equally likely outcomes are $\Omega = \{HH, HT, TH, TT\}$, where H means heads and T means tails. Let X count the heads:

$$X(HH) = 2, \qquad X(HT) = X(TH) = 1, \qquad X(TT) = 0.$$

To get exactly one head, we need either HT or TH. Therefore

$$X^{-1}(\{1\}) = \{HT, TH\}, \qquad P(X = 1) = P(\{HT, TH\}) = \frac{2}{4} = \frac{1}{2}.$$

Example 11: Counting tails in five coin tosses

Toss a fair coin five times, with independent tosses. Write H for heads and T for tails. The sample space is $\Omega = \{H, T\}^5$, containing $2^5 = 32$ sequences. Let X count the tails in a sequence, so its possible values are $0, 1, 2, 3, 4, 5$.

Each particular sequence has probability $(1/2)^5 = 1/32$. For example, $HHHHH$ and $TTHHH$ are equally likely, although they give different values of X.

No tails. Only $HHHHH$ gives zero tails:

$$\begin{aligned} P(X = 0) &= P\big(X^{-1}(\{0\})\big) \\ &= P(\{HHHHH\}) \\ &= \binom{5}{0}\left(\frac{1}{2}\right)^5 = \frac{1}{32}. \end{aligned}$$

One tail. The tail can occupy any of the five positions:

$$\begin{aligned} P(X = 1) &= P\big(X^{-1}(\{1\})\big) \\ &= P(\{THHHH, HTHHH, HHTHH, HHHTH, HHHHT\}) \\ &= \binom{5}{1}\left(\frac{1}{2}\right)^5 = \frac{5}{32}. \end{aligned}$$

Two tails. Choose which two of the five positions contain tails. The corresponding event contains $\binom{5}{2} = 10$ sequences:

$$\begin{aligned} P(X = 2) &= P\big(X^{-1}(\{2\})\big) \\ &= P(\{TTHHH, THTHH, \ldots, HHHTT\}) \\ &= \binom{5}{2}\left(\frac{1}{2}\right)^5 = \frac{10}{32}. \end{aligned}$$

In general, exactly x tails can occur in $\binom{5}{x}$ sequences. Summing their probabilities gives

$$\boxed{P(X = x) = P\big(X^{-1}(\{x\})\big) = \binom{5}{x}\left(\frac{1}{2}\right)^5}, \qquad x \in \{0, 1, 2, 3, 4, 5\},$$

The two factors are the number of sequences with x tails and the probability of each sequence.

Example 12: Darts: distance from the centre

Throw a dart at a circular target of radius $r > 0$. Assume that the dart always hits the disc and that its hitting point is uniform over the surface: regions with the same area have the same probability of being hit.

Place the centre at the origin. An outcome is the hitting point $\omega = (u, v)$, so the sample space is

$$\Omega = \{(u, v) \in \mathbb{R}^2 : u^2 + v^2 \leq r^2\}.$$

Let X measure the distance from that point to the centre:

$$X : \Omega \to \mathbb{R}, \qquad X(u, v) = \sqrt{u^2 + v^2}.$$

For a target of radius 10 cm, some possible outcomes and their distances are:

Hitting point ω	Distance $X(\omega)$ (cm)
$(0, 0)$	0
$(3, 4)$	$\sqrt{3^2 + 4^2} = 5$
$(-3, 4)$	$\sqrt{(-3)^2 + 4^2} = 5$
$(0, 10)$	10

In general, the possible values of X fill the interval $[0, r]$: zero at the centre, r on the boundary, and any distance in between.

For the 10 cm target, asking for a distance of at most 5 cm selects the outcomes

$$X^{-1}([0, 5]) = \{(u, v) \in \Omega : u^2 + v^2 \leq 25\}.$$

This is the smaller disc of radius 5 cm around the centre.

Now choose a distance threshold x. For $0 \leq x \leq r$, the event $X \leq x$ means that the dart hits the inner disc of radius x. Since the hitting point is uniform over the target, its probability is the ratio of the two areas:

$$P(X \leq x) = \frac{\text{area of the inner disc}}{\text{area of the target}} = \frac{\pi x^2}{\pi r^2} = \frac{x^2}{r^2}.$$

A negative threshold is impossible for a distance, while a threshold larger than r includes the entire target. Thus

$$\boxed{P(X \leq x) = \begin{cases} 0, & x < 0, \\ x^2/r^2, & 0 \leq x \leq r, \\ 1, & x > r. \end{cases}}$$

For the target of radius 10 cm,

$$P(X \leq 5) = \frac{\pi \cdot 5^2}{\pi \cdot 10^2} = \frac{25}{100} = \frac{1}{4}.$$

3.4 Discrete random variables

Definition 3.6: Discrete random variable and PMF

A random variable X is *discrete* if there is a finite or countably infinite set* $\mathcal{X} = \{x_1, x_2, \ldots\}$ such that $P(X \in \mathcal{X}) = 1$.

Its *probability mass function* (PMF) is

$$p_X(x) := P(X = x), \qquad x \in \mathbb{R}.$$

1. $p_X(x) \geq 0$ for every $x \in \mathbb{R}$.
2. The *support*, in the discrete sense used here, is $\text{supp}(X) := \{x \in \mathbb{R} : p_X(x) > 0\}$. Thus the support lists the values that have positive probability; values outside it have probability zero.
3. $\sum_{x \in \mathcal{X}} p_X(x) = 1$.

By countable additivity, for any set E,

$$P(X \in E) = \sum_{x \in E \cap \text{supp}(X)} p_X(x).$$

Example 13: The PMF of a fair die

Roll a fair six-sided die and let X be the number shown. The random variable X is discrete because the finite set $\mathcal{X} = \{1, 2, 3, 4, 5, 6\}$ satisfies $P(X \in \mathcal{X}) = 1$.

Its *probability mass function* (PMF) is:

$$p_X(x) = \begin{cases} 1/6, & x \in \{1, 2, 3, 4, 5, 6\}, \\ 0, & \text{otherwise.} \end{cases}$$

It satisfies $p_X(x) \geq 0$ for every x, and $\sum_{x \in \mathcal{X}} p_X(x) = 6 \cdot (1/6) = 1$.

Its *support* is $\text{supp}(X) = \{1, 2, 3, 4, 5, 6\}$, since each of these values has positive probability. For example, $p_X(3) = 1/6$, whereas $p_X(7) = 0$ and 7 is outside the support.

Let's see an example that shows how to use the PMF to calculate the probability of an event. To find the probability of rolling an even number, we add the probabilities of rolling 2, 4, or 6:

$$P(X \in \{2, 4, 6\}) = p_X(2) + p_X(4) + p_X(6) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}.$$

*A finite set has finitely many elements. A set A is countably infinite if there exists a bijection $f : \mathbb{N} \to A$.

3.5 Continuous random variables

For a continuous random variable, we describe how probability is concentrated along its possible values using a *probability density function* (PDF). The notation f_X names this function; the subscript X identifies the random variable it describes. For a possible value x, $f_X(x)$ is a number measuring the local concentration of probability: **probability per unit of length** near x. The definition below specifies precisely how this function represents the distribution of X.

Definition 3.7: Continuous random variable and PDF

A random variable X is *continuous* (in the sense used here) if there exists a function f_X, called its *probability density function* (PDF), such that, for every $a \leq b$,

$$P(a \leq X \leq b) = \int_a^b f_X(x)\,dx.$$

1. $f_X(x) \geq 0$ for every $x \in \mathbb{R}$.
2. Following the convention in the slides, the *support* is $\text{supp}(X) := \{x \in \mathbb{R} : f_X(x) > 0\}$.
3. $\int_{\mathbb{R}} f_X(x)\,dx = 1$, since the total area under f_X equals $P(X \in \mathbb{R})$, and X certainly takes some real value.

To make the meaning of $f_X(x)$ explicit, fix a value x and consider an interval centred at x, with positive width h:

$$\left[x - \frac{h}{2},\ x + \frac{h}{2}\right].$$

Its probability is $P(x - h/2 \leq X \leq x + h/2)$. Dividing this probability by h gives the **average probability per unit of length** over that interval. Keep x fixed and shrink the interval by letting $h \to 0^+$. If f_X is continuous at x, then

$$\boxed{f_X(x) = \lim_{h \to 0^+} \frac{P(x - h/2 \leq X \leq x + h/2)}{h}.}$$

Indeed, by the definition above, the ratio is

$$\frac{1}{h} \int_{x-h/2}^{x+h/2} f_X(t)\,dt,$$

the average value of the PDF on that interval. Continuity at x ensures that this average approaches $f_X(x)$ as the interval shrinks. This limit is a local interpretation of the PDF under that continuity assumption; the integral definition above is the general one.

Note: Heads-up

A PDF cannot concentrate probability on a countable set. First, setting $a = b = x_i$ in the definition gives

$$P(X = x_i) = \int_{x_i}^{x_i} f_X(t)\,dt = 0 :$$

an interval of zero length has zero area under the PDF.

Now let $A = \{x_1, x_2, \ldots\}$ be any finite or countably infinite set, listing its elements without repetitions. The event $X \in A$ is the union of the pairwise disjoint events $\{X = x_i\}$. By countable additivity,

$$P(X \in A) = \sum_i P(X = x_i) = \sum_i 0 = 0.$$

Thus no finite or countably infinite set can contain X with probability 1. The existence of a PDF already implies this; having an uncountable set of possible values alone does not guarantee that a PDF exists.

Example 14: Waiting for a bus

Suppose a bus arrives every 10 minutes and you reach the stop at a uniformly random time between two arrivals. Let X be your waiting time in minutes. Its possible values lie in $[0, 10]$. Uniformity means that the probability of a waiting-time interval is its length divided by the full length, 10 minutes.

For example, fix $x = 5$ minutes and consider a small interval centred at 5, of width h minutes. As long as the interval stays inside $[0, 10]$, its probability is $h/10$. Therefore,

$$\frac{P(5 - h/2 \le X \le 5 + h/2)}{h} = \frac{h/10}{h} = \frac{1}{10}.$$

For widths $h = 1$, 0.2 and 0.02, the probabilities are 0.1, 0.02 and 0.002: the interval's probability decreases, but the probability per minute remains $1/10$. The same calculation works at every interior point $0 < x < 10$, so the PDF is constant there, with value $1/10$.

Choosing the same value at the endpoints, its PDF is:

$$f_X(x) = \begin{cases} 1/10, & 0 \le x \le 10, \\ 0, & \text{otherwise.} \end{cases}$$

It satisfies $f_X(x) \ge 0$ for every x, and $\int_{\mathbb{R}} f_X(x)\,dx = \int_0^{10} (1/10)\,dx = 1$.

Its *support*, using the convention above, is $[0, 10]$.

Let's see an example that shows how to use the PDF to calculate the probability of an event. The probability of waiting between 2 and 5 minutes is:

$$P(2 \le X \le 5) = \int_2^5 \frac{1}{10}\,dx = \frac{5 - 2}{10} = 0.30.$$

3.6 Cumulative distribution function (CDF)

Definition 3.8: Cumulative distribution function

Given a random variable X, its *cumulative distribution function* (CDF) is the function $F_X : \mathbb{R} \to [0,1]$ defined by

$$F_X(x) := P(X \leq x) = P\big(X \in (-\infty, x]\big), \qquad x \in \mathbb{R}.$$

It gives the probability accumulated up to and including the threshold x. It is defined for both discrete and continuous random variables.

Proposition: The CDF determines the distribution

The CDF contains all the information about the *distribution* of a random variable. If $F_X(t) = F_Y(t)$ for every $t \in \mathbb{R}$, then X and Y have the same distribution: for any set of values E defining an event,

$$P(X \in E) = P(Y \in E).$$

This does not mean that X and Y must produce the same outcome in an experiment.

Proof. We prove only a special case of this statement: we show that it holds for intervals $E = (a, b]$. The full result also holds for any set E defining an event, but we do not prove that general case here.

Suppose $F_X(t) = F_Y(t)$ for every $t \in \mathbb{R}$, and let $a < b$. Since

$$\{X \leq b\} = \{X \leq a\} \cup \{a < X \leq b\},$$

and the two events on the right are disjoint, additivity gives

$$P(a < X \leq b) = P(X \leq b) - P(X \leq a) = F_X(b) - F_X(a).$$

Applying the same reasoning to Y and using the equality of the two CDFs, we obtain

$$\begin{aligned} P(X \in E) &= F_X(b) - F_X(a) \\ &= F_Y(b) - F_Y(a) \\ &= P(Y \in E). \end{aligned}$$

□

Note: PMF and CDF

The PMF answers: *"What is the probability that X is exactly x?"*

$$p_X(x) = P(X = x).$$

The CDF answers: *"What is the probability that X is at most x?"*

$$F_X(x) = P(X \leq x).$$

3.6.1 Properties of the CDF

Proposition 3.1: Properties of the CDF (I)

A function $F(\cdot)$ mapping the real line to $[0, 1]$ is a CDF for some probability measure $P(\cdot)$ if and only if:

1. $F(\cdot)$ is *non-decreasing*: $x_1 < x_2$ implies that $F(x_1) \leq F(x_2)$.
2. $F(\cdot)$ is *normalized*:
$$\lim_{x \to -\infty} F(x) = 0, \qquad \lim_{x \to +\infty} F(x) = 1.$$
3. $F(\cdot)$ is *right-continuous*: for all x we have
$$F(x) = F(x^+), \qquad F(x^+) := \lim_{t \downarrow x} F(t).$$

Proposition 3.2: Properties of the CDF (II/a)

Let $F_X(\cdot)$ be the CDF of a random variable X. Then:

1. For $x_1 < x_2$,
$$P(x_1 < X \leq x_2) = P\big(X \in (x_1, x_2]\big) = F_X(x_2) - F_X(x_1).$$
In the discrete case, be careful with the strict and non-strict inequalities.
2.
$$P(X > x) = P\big(X \in (x, +\infty)\big) = 1 - F_X(x).$$
3. If X is continuous with PDF $f_X(\cdot)$, then
$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt.$$
A PDF can be recovered from the CDF as
$$f_X(x) = F_X'(x)$$
at points where F_X is differentiable.

Proposition 3.3: Properties of the CDF (II/b)

Let $F_X(\cdot)$ be the CDF of a random variable X. Then:

1. $$P(x_1 < X \leq x_2) = P\big(X \in (x_1, x_2]\big) = F_X(x_2) - F_X(x_1), \qquad x_1 < x_2.$$
2. $$P(X > x) = P\big(X \in (x, +\infty)\big) = 1 - F_X(x).$$
3. If X is discrete with PMF $p_X(\cdot)$ and support $\mathcal{X} = \{x_1, x_2, \ldots\}$, then
$$F_X(x) = \sum_{x_i \leq x} p_X(x_i) = P(X < x) + p_X(x).$$
Hence
$$p_X(x) = F_X(x) - F_X(x^-), \qquad F_X(x^-) := \lim_{t \uparrow x} F_X(t).$$

Let $F_X(\cdot)$ be the CDF of a continuous random variable with PDF $f_X(\cdot)$. The following informal observations motivate later nonparametric statistical developments:

1. Roughly speaking, $F_X(\cdot)$ is "simpler" than $f_X(\cdot)$.
2. It must be non-decreasing, so its shape is more constrained than that of a PDF.
3. Integration accumulates and smooths the PDF, making the CDF more regular.
4. This suggests that estimating the CDF is easier than estimating the PDF: fewer observations may be needed to reach the same statistical accuracy.

Example 15: Five coin tosses: PMF and CDF

Return to five independent flips of a balanced coin, and let X count the tails. Its possible values are $0, 1, 2, 3, 4, 5$.

Its *probability mass function* (PMF) is:

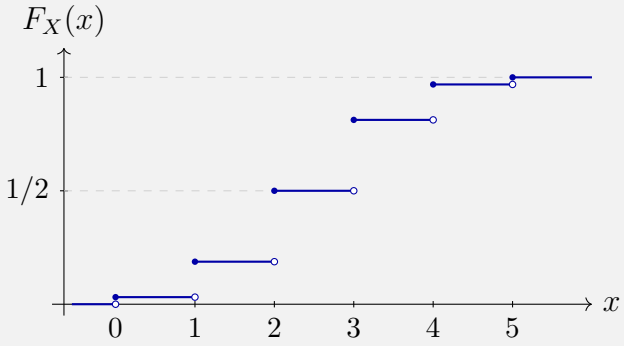
$$p_X(x) = P(X = x) = \binom{5}{x}\left(\frac{1}{2}\right)^5, \qquad x \in \{0, 1, 2, 3, 4, 5\},$$

and $p_X(x) = 0$ otherwise. The probabilities are:

x	0	1	2	3	4	5
$p_X(x)$	1/32	5/32	10/32	10/32	5/32	1/32

Its *cumulative distribution function* (CDF) is obtained by adding the probabilities of all values at most x:

$$F_X(x) = \sum_{k \le x} p_X(k) = \begin{cases} 0, & x < 0, \\ 1/32, & 0 \le x < 1, \\ 6/32, & 1 \le x < 2, \\ 16/32, & 2 \le x < 3, \\ 26/32, & 3 \le x < 4, \\ 31/32, & 4 \le x < 5, \\ 1, & x \ge 5. \end{cases}$$



Filled points give the value of the CDF at each jump; open points are excluded.

The graph displays the three properties in *Properties of the CDF (I)*:

1. *Non-decreasing*: each jump adds a positive probability.
2. *Normalized*: $F_X(x) = 0$ for $x < 0$ and $F_X(x) = 1$ for $x \ge 5$.
3. *Right-continuous*: for example, consider $x = 2$. We have $F_X(2) = 16/32 = 1/2$, and the same value holds for $2 \le x < 3$. Thus the right limit equals $F_X(2)$.

Example 16: Darts: from areas to the CDF and PDF

Return to the earlier darts example, with a disc of radius 10 cm and a hitting point uniform over its area. Let X be the distance from the centre in centimetres. Earlier, we obtained $P(X \le x)$ using areas. Now we recognise this function as the CDF, differentiate it to find a PDF, and use that PDF to calculate probabilities of intervals.

1. Recognise the CDF. The function calculated earlier is the CDF:

$$F_X(x) = P(X \le x) = \begin{cases} 0, & x < 0, \\ x^2/100, & 0 \le x \le 10, \\ 1, & x > 10. \end{cases}$$

2. Derive the PDF. The PDF is not given. We find it by differentiating the CDF for $0 < x < 10$:

$$f_X(x) = F_X'(x) = \frac{d}{dx}\left(\frac{x^2}{100}\right) = \frac{2x}{100} = \frac{x}{50}.$$

Its PDF is:

$$f_X(x) = \begin{cases} x/50, & 0 \le x \le 10, \\ 0, & \text{otherwise.} \end{cases}$$

This PDF reproduces the probabilities of the model. It is non-negative and satisfies

$$\int_{\mathbb{R}} f_X(x)\,dx = \int_0^{10} \frac{x}{50}\,dx = \left[\frac{x^2}{100}\right]_0^{10} = 1.$$

Its support, with the convention used above and this choice of PDF, is $(0, 10]$.

3. Use the PDF to calculate an interval probability. Now that we have derived the PDF, we can, for example, calculate the probability that the dart lands between 5 and 8 cm from the centre:

$$P(5 \le X \le 8) = \int_5^8 \frac{x}{50}\,dx = \left[\frac{x^2}{100}\right]_5^8 = \frac{64 - 25}{100} = 0.39.$$

The CDF gives the same result: $F_X(8) - F_X(5) = 0.64 - 0.25 = 0.39$.

For the two intervals used in the slides, the same method gives

$$P(0 < X \le 5) = \int_0^5 \frac{x}{50}\,dx = \frac{1}{4}, \qquad P(5 < X \le 10) = \int_5^{10} \frac{x}{50}\,dx = \frac{3}{4}.$$

3.7 Relations

3.7.1 From the PMF to the CDF

What we are trying to answer

If we know the probability of each individual value of X, how do we calculate the probability that X is at most x?

For a discrete random variable, the event $X \leq x$ includes every possible value k that is at most x. These events $\{X = k\}$ are pairwise disjoint, so we add their probabilities:

$$\underbrace{F_X(x)}_{\text{CDF: to find}} = P(X \leq x) = \sum_{\substack{k \in \text{supp}(X) \\ k \leq x}} \underbrace{p_X(k)}_{\text{PMF: known}}, \qquad x \in \mathbb{R}.$$

3.7.2 From the PDF to the CDF

What we are trying to answer

We know how probability is spread over the possible values of X (its PDF). How do we calculate the probability of getting a value less than or equal to x?

For a continuous random variable with PDF f_X, we calculate $P(X \leq x)$ by integrating the PDF over $(-\infty, x]$:

$$\underbrace{F_X(x)}_{\text{CDF: to find}} = P(X \leq x) = \int_{-\infty}^{x} \underbrace{f_X(t)}_{\text{PDF: known}} \, dt, \qquad x \in \mathbb{R}.$$

3.7.3 From the CDF to the PMF and PDF

What we are trying to answer

We know the CDF F_X. How do we recover the PMF p_X if X is discrete, or the PDF f_X if X is continuous?

What we know. The CDF is defined by

$$F_X(x) := P(X \leq x).$$

For every threshold x, it gives the probability of obtaining a value less than or equal to x. We want to recover the PMF or the PDF, depending on the type of X.

Discrete case: recover the PMF. The PMF is defined by

$$p_X(x) := P(X = x).$$

The CDF includes all values up to x. To isolate the probability of the single value x, we subtract the probability of values strictly less than x:

$$P(X = x) = P(X \leq x) - P(X < x).$$

Since the left limit satisfies

$$F_X(x^-) := \lim_{t \uparrow x} F_X(t) = P(X < x),$$

we obtain

$$\boxed{\underbrace{p_X(x)}_{\text{PMF: to find}} = \underbrace{F_X(x) - F_X(x^-)}_{\text{CDF: known}}.}$$

Thus the PMF at x is the size of the jump of the CDF at x.

Continuous case: recover the PDF. A PDF is a non-negative function f_X such that

$$P(a \leq X \leq b) = \int_a^b f_X(t)\,dt, \qquad \int_{\mathbb{R}} f_X(t)\,dt = 1.$$

In particular,

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)\,dt.$$

The CDF is therefore the integral of the PDF. To recover a PDF, we differentiate the CDF:

$$\boxed{\underbrace{f_X(x)}_{\text{PDF: to find}} = \frac{d}{dx} \underbrace{F_X(x)}_{\text{CDF: known}}.}$$

This recovers a PDF at points where the derivative exists, by the fundamental theorem of calculus.

3.8 The quantile function

What we are trying to answer

Given a probability p, which threshold x satisfies $P(X \leq x) = p$?

The CDF starts with a threshold and returns a probability. The quantile function starts with a probability and returns a threshold. For example, the 90th percentile is a threshold below which 90% of the probability lies; the median is the quantile at level 0.5.

Definition 3.9: Quantile function: continuous case

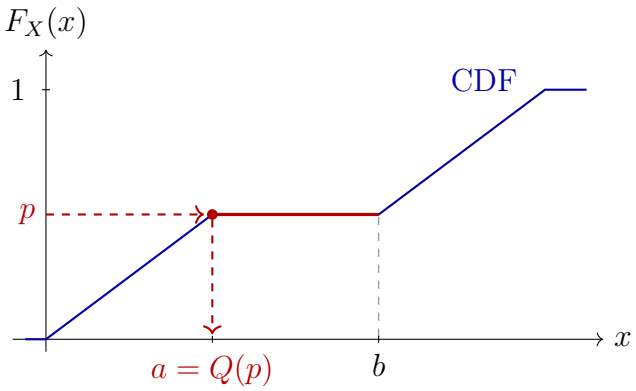
Let X be a continuous random variable with CDF F_X. For a probability level $0 < p < 1$, its *quantile function* is

$$Q(p) := \inf\{x \in \mathbb{R} : F_X(x) = p\}.$$

It gives the smallest value x such that $F_X(x) = P(X \leq x) = p$. We also write q_p instead of $Q(p)$.

The symbol inf denotes the infimum (greatest lower bound); when the set has a minimum, the infimum is its smallest element. Here we consider levels strictly between 0 and 1; endpoint levels may require separate conventions.

Geometric interpretation. Start at height p on the vertical axis, move horizontally to the CDF, and then read the corresponding threshold on the horizontal axis. If several thresholds give the same probability, choose the smallest one.



In this continuous CDF, every threshold in $[a, b]$ gives the same probability p. Thus $\{x : F_X(x) = p\} = [a, b]$ and $Q(p) = a$.

When the CDF is strictly increasing between probability levels 0 and 1, each level corresponds to a unique threshold. The CDF is therefore invertible on this range, so

$$\boxed{Q(p) = F_X^{-1}(p).}$$

An inverse may exist without a simple explicit formula. In that case, quantiles can be calculated numerically.

A more general definition. If the CDF has a jump, it may skip the level p, so the equation $F_X(x) = p$ may have no solution. We therefore replace equality with $F_X(x) \geq p$.

Definition 3.10: Quantile function: general case

Let X be any random variable, discrete or continuous, with CDF F_X. For a probability level $0 < p < 1$, its *quantile function* is

$$Q(p) := \inf\{x \in \mathbb{R} : F_X(x) \geq p\}.$$

It gives the smallest threshold x such that $P(X \leq x)$ is at least p. We also write q_p instead of $Q(p)$.

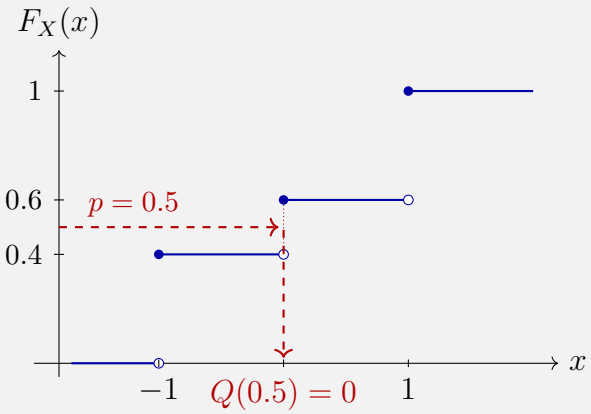
At a jump, the quantile is the first threshold where the CDF reaches or exceeds p; the probability there may be greater than p. If the CDF has a plateau at level p, this definition chooses its left endpoint. This is the lower-quantile convention used here. If the CDF is invertible, $Q(p) = F_X^{-1}(p)$ (see Definition 3.9).

Example 17: Reading a quantile across a jump

Let X take the values $-1, 0, 1$ with probabilities $0.4, 0.2, 0.4$, respectively. Its CDF is

$$F_X(x) = \begin{cases} 0, & x < -1, \\ 0.4, & -1 \leq x < 0, \\ 0.6, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

We want the median, $Q(0.5)$.



Start at height 0.5 and move to the right. Before $x = 0$, the CDF is below 0.5. At $x = 0$, the filled point gives $F_X(0) = 0.6 \geq 0.5$: this is the first threshold that qualifies. Therefore,

$$\{x : F_X(x) \geq 0.5\} = [0, +\infty), \qquad \boxed{Q(0.5) = \inf[0, +\infty) = 0.}$$

Here $P(X \leq Q(0.5)) = 0.6$, which exceeds the requested level 0.5.

Note: Quantiles as areas under the PDF

What we are trying to answer. We know the PDF and choose a probability p. Where should we stop so that the area under the PDF to the left is p?

For a continuous random variable with a PDF, we already know the relation between the PDF and the CDF:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)\, dt.$$

Previously, we fixed the threshold x and calculated the probability. Now we fix the probability p and look for the threshold $q_p = Q(p)$. Since the CDF is continuous, $F_X(q_p) = p$, so

$$\underbrace{p}_{\text{chosen probability}} = F_X(q_p) = \int_{-\infty}^{\overbrace{q_p}^{\text{threshold: to find}}} \underbrace{f_X(t)}_{\text{PDF: known}}\, dt.$$

Thus we find q_p by solving this equation. If several thresholds give the same area, the general quantile definition chooses the smallest one.

For the median, $p = 0.5$: we look for the threshold that leaves half of the total area under the PDF to its left.

Example 18: Quantiles of a Pareto distribution

The slides give a continuous random variable with CDF

$$F_X(x) = \begin{cases} 0, & x < 1, \\ 1 - x^{-\alpha}, & x \geq 1, \end{cases} \qquad \alpha > 0.$$

This is a Pareto distribution with shape parameter α and scale parameter 1. For example, if $\alpha = 1/2$, suppose we want to find the 20th percentile and the median.

1. Recover the PDF. For $x > 1$, we differentiate the given CDF:

$$f_X(x) = F_X'(x) = \alpha x^{-\alpha-1}.$$

We may therefore choose

$$f_X(x) = \begin{cases} \alpha/x^{\alpha+1}, & x \geq 1, \\ 0, & x < 1. \end{cases}$$

2. Find the quantile function. The CDF is strictly increasing for $x \geq 1$. For $0 < p < 1$, we find the unique threshold q_p by solving $F_X(q_p) = p$:

$$\begin{aligned} p &= 1 - q_p^{-\alpha}, \\ q_p^{-\alpha} &= 1 - p, \\ q_p^{\alpha} &= \frac{1}{1-p}, \\ q_p &= \left(\frac{1}{1-p}\right)^{1/\alpha}. \end{aligned}$$

Thus the whole quantile function is

$$\boxed{Q(p) = F_X^{-1}(p) = (1-p)^{-1/\alpha}, \qquad 0 < p < 1.}$$

3. Calculate the requested quantiles with $\alpha = 1/2$. Here $1/\alpha = 2$, so $Q(p) = (1-p)^{-2}$. For the 20th percentile, we choose $p = 0.20$:

$$\boxed{q_{0.20} = (1 - 0.20)^{-2} = \frac{1}{0.8^2} = 1.5625 \approx 1.56.}$$

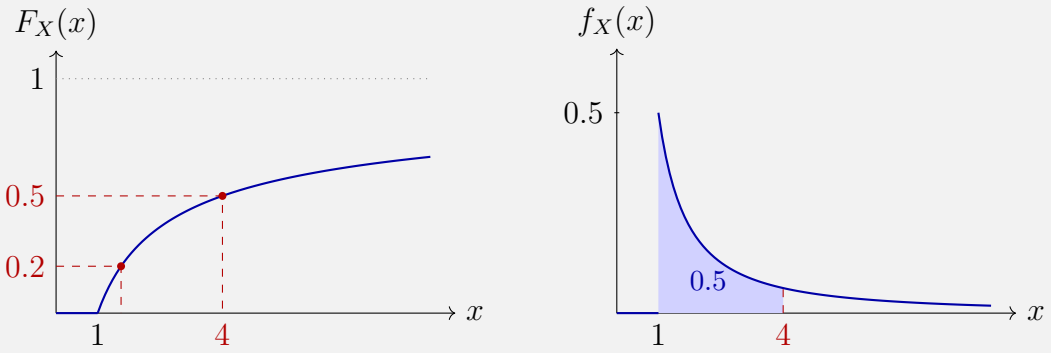
Hence $P(X \leq 1.5625) = 0.20$. For the median (the second quartile), we choose $p = 0.50$:

$$\boxed{q_{0.50} = (1 - 0.50)^{-2} = \frac{1}{0.5^2} = 4.}$$

Hence $P(X \leq 4) = 0.50$: the threshold 4 leaves half of the probability to its left.

Note: Reading the Pareto quantiles geometrically

For the Pareto example with $\alpha = 1/2$, the two graphs show the same median. On the CDF, height 0.5 corresponds to $x = 4$. Under the PDF, the area from 1 to 4 is 0.5.



The area calculation confirms the median:

$$\int_1^4 \frac{1}{2t^{3/2}}\,dt = \left[-\frac{1}{\sqrt{t}}\right]_1^4 = -\frac{1}{2} + 1 = \frac{1}{2}.$$

Example 19: Quantiles of a discrete distribution

The second slide example uses the same variable as the earlier geometric example:

x	-1	0	1
$p_X(x)$	$2/5$	$1/5$	$2/5$

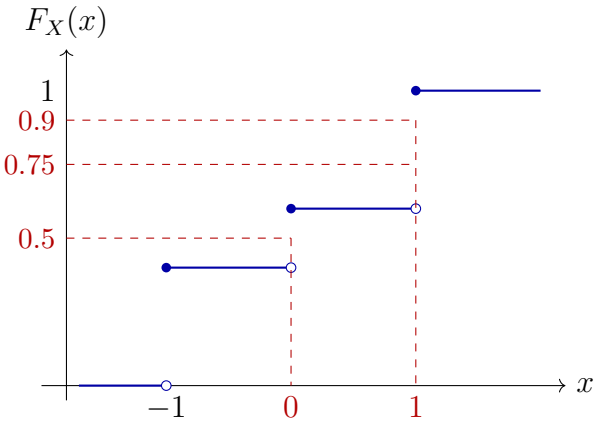
We want the median, the 90th percentile, and the third quartile.

1. Build the CDF by adding the masses. At -1 we have $2/5$; at 0 we have $2/5 + 1/5 = 3/5$; at 1 we have $2/5 + 1/5 + 2/5 = 1$. Therefore,

$$F_X(x) = \begin{cases} 0, & x < -1, \\ 2/5 = 0.4, & -1 \le x < 0, \\ 3/5 = 0.6, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

2. Find the first threshold whose CDF reaches each level. We use $q_p = \inf\{x : F_X(x) \ge p\}$.

- **Median:** $p = 0.50$**.** For $x < 0$, $F_X(x) \le 0.4 < 0.5$, whereas $F_X(0) = 0.6 \ge 0.5$. Thus
$$\{x : F_X(x) \ge 0.5\} = [0, +\infty), \qquad \boxed{q_{0.50} = 0.}$$
- 90**th percentile:** $p = 0.90$**.** For $x < 1$, $F_X(x) \le 0.6 < 0.9$, whereas $F_X(1) = 1 \ge 0.9$. Thus
$$\{x : F_X(x) \ge 0.9\} = [1, +\infty), \qquad \boxed{q_{0.90} = 1.}$$
- **Third quartile:** $p = 0.75$**.** For $x < 1$, $F_X(x) \le 0.6 < 0.75$, whereas $F_X(1) = 1 \ge 0.75$. Thus
$$\{x : F_X(x) \ge 0.75\} = [1, +\infty), \qquad \boxed{q_{0.75} = 1.}$$



The median is 0, while both the 90th percentile and the third quartile are 1: different probability levels can give the same quantile when the CDF has jumps.

3.9 Common univariate distributions

We collect the univariate distribution families here. Their later uses in joint models and simulations refer to these definitions. In the formulas, the **indicator function** of a set A is

$$\mathbf{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Multiplying a density by this indicator makes it zero outside A.

3.9.1 The uniform distribution

A **uniform distribution** assigns the same probability to intervals of the same length within its support. Its parameters a and b specify the ends of that support.

Definition 3.11: Uniform distribution

A continuous random variable X has a *uniform distribution* with parameters $a < b$, where $a, b \in \mathbb{R}$, if its PDF and CDF are

$$f_X(x) = \frac{1}{b-a}\mathbf{1}_{[a,b]}(x),$$

$$F_X(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \le x \le b, \\ 1, & x > b. \end{cases}$$

We denote this distribution by $\text{Unif}(a, b)$ and write $X \sim \text{Unif}(a, b)$. Here Unif abbreviates *uniform*, and $\sim$ means *has distribution*.

Why is the density a rectangle? Uniformity means that intervals of the same length within $[a, b]$ have the same probability, hence the same area under the PDF. If the density were higher in one region, a small interval there would have more probability than an equally long interval in a lower region. The density is therefore constant on $[a, b]$ and zero outside it. Its graph is horizontal on the support, so the region beneath it is a rectangle: the rectangular shape follows from uniformity, before we calculate its height.

Why is the height $1/(b-a)$? Let h be the constant height. The rectangle has width $b - a$, and the total area under a PDF must be 1:

$$h(b-a) = 1 \qquad \Longrightarrow \qquad h = \frac{1}{b-a}.$$

For $a \le c \le d \le b$,

$$P(c \le X \le d) = \int_c^d \frac{1}{b-a}\,dx = \frac{d-c}{b-a}.$$

The numerator is the length of the selected interval; the denominator is the length of the full support.

For $a \leq x \leq b$, the same calculation gives the CDF:

$$F_X(x) = \int_a^x \frac{1}{b-a}\, dt = \frac{x-a}{b-a}.$$

3.9.1.1 The standard uniform distribution: Unif$(0,1)$

The **standard uniform distribution**, written $U \sim \text{Unif}(0,1)$, is the **special case** of the general Unif(a,b) distribution in Section 3.9.1 obtained by setting $a = 0$ and $b = 1$.
Its support is therefore $[0,1]$, and substituting these endpoints gives the constant density $1/(b-a) = 1/(1-0) = 1$. Its PDF is

$$f_U(u) = \begin{cases} 1, & 0 \leq u \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

For $0 \leq c \leq d \leq 1$,

$$P(c \leq U \leq d) = \int_c^d 1\, du = d - c.$$

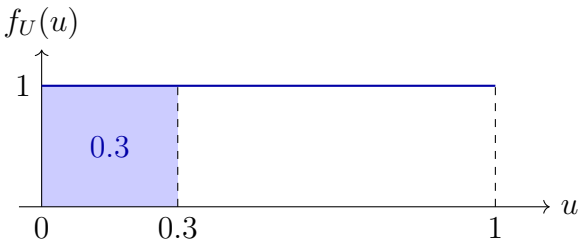
Find the CDF. By the definition of a CDF, $F_U(t) = P(U \leq t)$. For $0 \leq t \leq 1$, integrate the PDF up to t:

$$\boxed{F_U(t) = P(U \leq t) = \int_0^t 1\, du = t.}$$

This also follows directly from the general CDF formula: $(t-a)/(b-a) = (t-0)/(1-0) = t$ for $0 \leq t \leq 1$. The full CDF is therefore

$$F_U(t) = \begin{cases} 0, & t < 0, \\ t, & 0 \leq t \leq 1, \\ 1, & t > 1. \end{cases}$$

For example, $P(U \leq 0.3) = 0.3$: the interval from 0 to 0.3 contains 30% of the probability. Geometrically, this is a rectangle with width 0.3 and height 1, so its area is 0.3.



3.9.2 The exponential distribution as a waiting-time model

Let T be the time until the next event, such as a job reaching a printer. Times are nonnegative. The parameter $\lambda > 0$ is a **rate**, measured in inverse time units: a larger rate describes shorter waits.

Definition 3.12: Exponential distribution

We write $T \sim \mathrm{Exp}(\lambda)$, with rate $\lambda > 0$, when its PDF and CDF are

$$f_T(t) = \lambda e^{-\lambda t} \mathbf{1}_{[0,+\infty)}(t), \qquad F_T(t) = \begin{cases} 0, & t < 0, \\ 1 - e^{-\lambda t}, & t \geq 0. \end{cases}$$

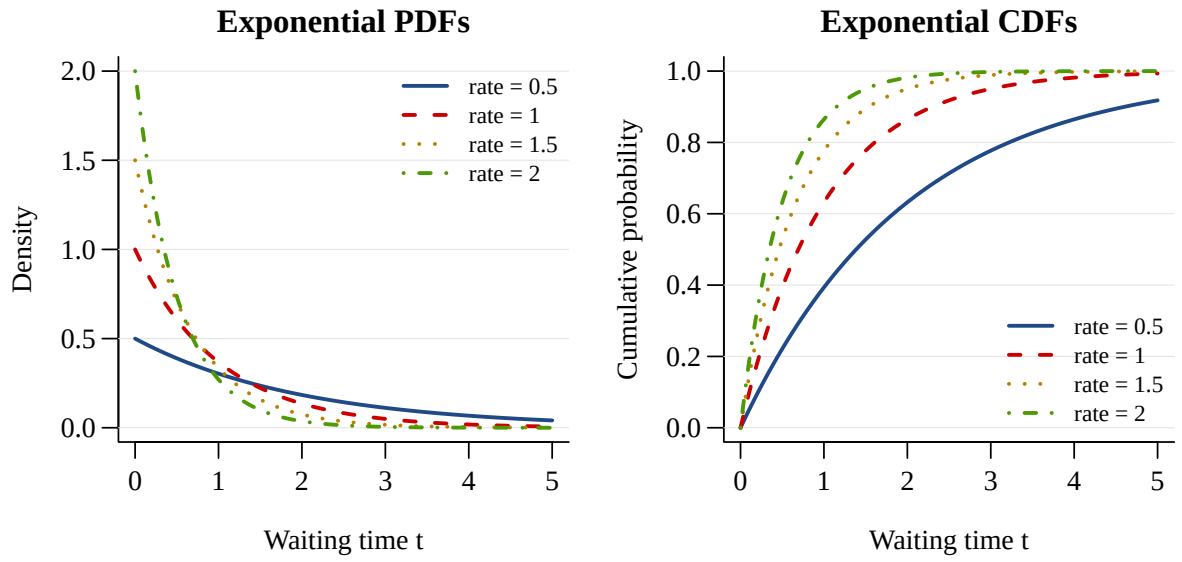
The density integrates to 1 because

$$\int_0^{+\infty} \lambda e^{-\lambda t}\, dt = \left[-e^{-\lambda t}\right]_0^{+\infty} = 1.$$

Integrating only up to a threshold $t \geq 0$ gives the CDF. Consequently, the probability of waiting longer than t is

$$P(T > t) = 1 - F_T(t) = e^{-\lambda t}.$$

This is the **survival function**.



The left panel compares PDFs for four rates. A larger λ places more density near zero, and the tail decays faster. The right panel shows the corresponding CDFs: with a larger rate, more probability accumulates before the same waiting-time threshold. Every PDF still has total area 1.

3.9.2.1 *A constant chance of arrival over a short interval*

Suppose no event has occurred by time t. For a short additional interval of length $h > 0$, the model gives

$$\begin{aligned} P(t < T \leq t + h \mid T > t) &= \frac{F_T(t+h) - F_T(t)}{P(T > t)} \\ &= \frac{e^{-\lambda t} - e^{-\lambda(t+h)}}{e^{-\lambda t}} \\ &= 1 - e^{-\lambda h} \approx \lambda h. \end{aligned}$$

The approximation applies when λh is small. The exact probability depends on the interval's length, not on how long we have already waited. Equivalently, the instantaneous event rate is constant. An average arrival rate alone does not guarantee an exponential waiting time; this constant-rate assumption is part of the model.

Example 20: The next printer job

Jobs reach a printer at an average rate of 3 per hour. Assume the waiting time follows $T \sim \text{Exp}(3)$, with T measured in hours. What is the probability that the next job arrives within 5 minutes?

Convert the time, then evaluate the CDF.

Step 1: Use the same time unit as the rate. Five minutes is $5/60 = 1/12$ of an hour.

Step 2: Translate the question into an event. We want $P(T \leq 1/12) = F_T(1/12)$.

Step 3: Substitute the rate and threshold.

$$\boxed{P(T \leq 1/12) = 1 - e^{-3/12} = 1 - e^{-1/4} \approx 0.2212.}$$

Under this model, the probability is about 22.12%.

3.9.2.2 *The memoryless property*

If we have already waited t time units, how long must we wait from now? For $s, t \geq 0$,

$$\begin{aligned} P(T > t + s \mid T > t) &= \frac{P(T > t+s)}{P(T > t)} \\ &= \frac{e^{-\lambda(t+s)}}{e^{-\lambda t}} \\ &= e^{-\lambda s} = P(T > s). \end{aligned}$$

This is called **memorylessness**: conditional on the event not having occurred, the remaining waiting time has the same exponential distribution as the original waiting time. It is a statement about the model's conditional distribution.

3.9.2.3 The PDF and CDF in R

R evaluates these two functions with `dexp` and `pexp`. For example, the following commands plot the rate-1 PDF and CDF; changing `rate` gives the curves above.

```
curve(dexp(x, rate = 1), from = 0, to = 5,
      xlab = "Waiting time", ylab = "Density")
curve(pexp(x, rate = 1), from = 0, to = 5,
      xlab = "Waiting time", ylab = "Probability")
pexp(5 / 60, rate = 3)  # 0.2211992
```

3.9.3 The Bernoulli distribution

One trial with a success or failure can be recorded as 1 or 0.

Definition 3.13: Bernoulli distribution

For $0 \leq p \leq 1$, we write $X \sim \text{Ber}(p)$ when

$$p_X(x) = \begin{cases} 1-p, & x = 0, \\ p, & x = 1, \\ 0, & \text{otherwise.} \end{cases}$$

The parameter p is the success probability. The two masses add to 1.

3.9.4 The binomial distribution

Repeat a Bernoulli trial independently n times, with the same success probability p. Count the successes rather than record their order.

Definition 3.14: Binomial distribution

For an integer $n \geq 1$ and $0 < p < 1$, $X \sim \text{Bin}(n, p)$ has PMF

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, \ldots, n,$$

and zero elsewhere. For $p = 0$, $X = 0$ with probability 1; for $p = 1$, $X = n$ with probability 1.

For $0 < p < 1$, each ordered sequence with k successes has probability $p^k(1-p)^{n-k}$. There are $\binom{n}{k}$ such sequences, which explains the PMF. The binomial theorem gives total mass $(p + (1-p))^n = 1$.

3.9.5 The Beta distribution

Let P take values in $(0, 1)$, and write $\pi(p)$ for its PDF.

Definition 3.15: Beta distribution

For $\alpha, \beta > 0$, the density of $P \sim \text{Beta}(\alpha, \beta)$ is

$$\pi(p) = \frac{p^{\alpha-1}(1-p)^{\beta-1}}{B(\alpha, \beta)}, \qquad 0 < p < 1,$$

and it is zero elsewhere. The denominator is the normalizing constant

$$B(\alpha, \beta) = \int_0^1 t^{\alpha-1}(1-t)^{\beta-1}\, dt.$$

To normalize this density, divide $p^{\alpha-1}(1-p)^{\beta-1}$ by its total integral. Positivity of α and β makes this integral finite, including when the density grows without bound near an endpoint. Such a peak is not a point mass; $\Pr(P = 0) = \Pr(P = 1) = 0$.

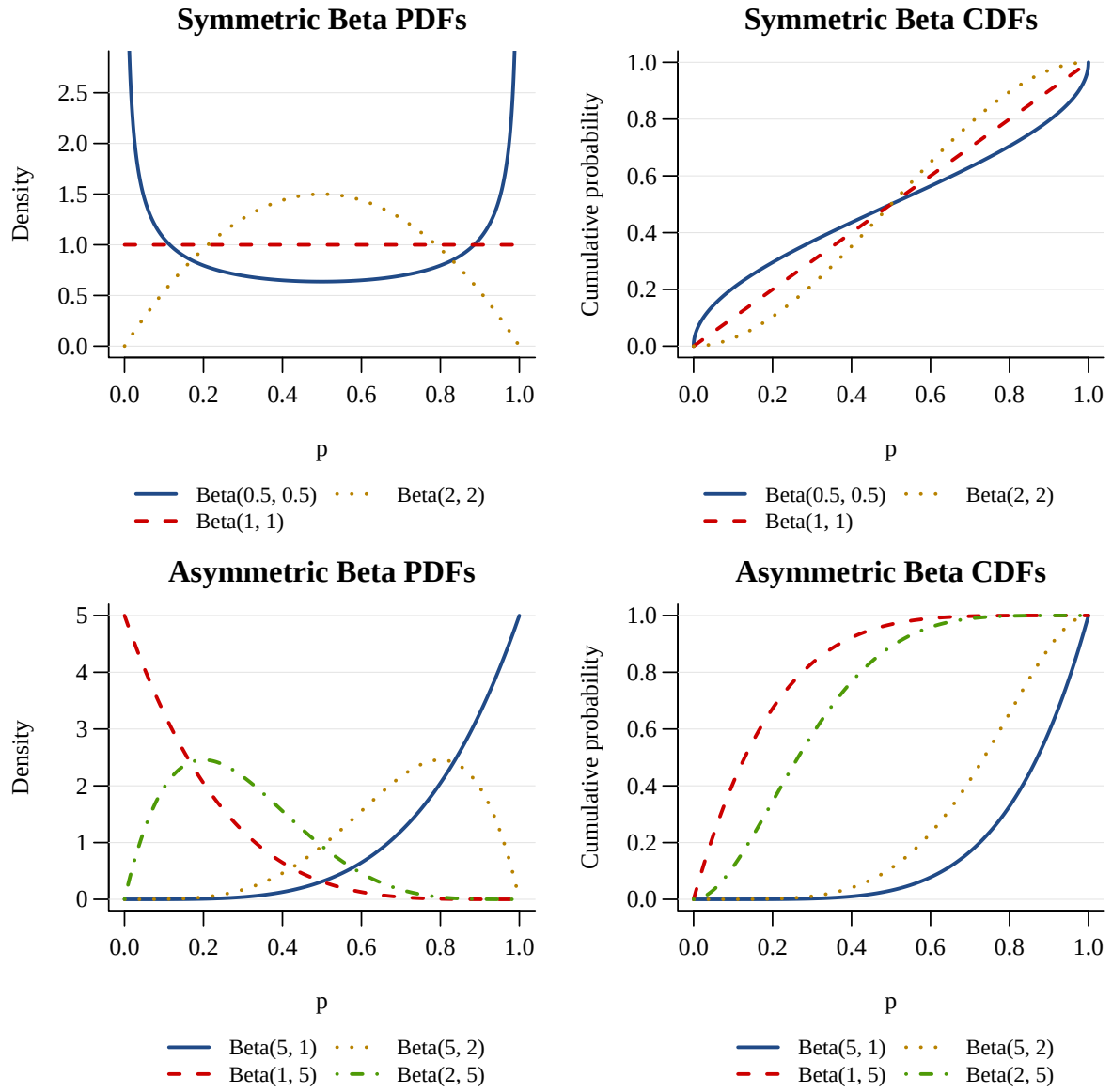
For a threshold $0 \le u \le 1$, the CDF accumulates only the part of the numerator to the left of u:

$$F_P(u) = \frac{B(u; \alpha, \beta)}{B(\alpha, \beta)}, \qquad B(u; \alpha, \beta) = \int_0^u t^{\alpha-1}(1-t)^{\beta-1}\, dt.$$

The numerator is the **incomplete Beta function**; the CDF is 0 for $u < 0$ and 1 for $u > 1$. The full Beta function also satisfies

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}, \qquad \Gamma(a) = \int_0^{+\infty} t^{a-1}e^{-t}\, dt, \quad a > 0.$$

Here Γ names the Gamma function. For general real parameters, we usually evaluate these functions numerically.



The parameters α, β control the shape:

- With $\alpha = \beta = 1$, the numerator and its integral are both 1, so the distribution is $\text{Unif}(0, 1)$.
- With both parameters below 1, the density has peaks near 0 and 1. The plot for $(0.5, 0.5)$ cuts off their unbounded heights to make the rest of the curve visible.
- With both parameters above 1, the density has one interior peak. The symmetric example $(2, 2)$ concentrates probability around $1/2$.
- When $\alpha = \beta$, the density is symmetric about $1/2$. The examples $(5, 1)$ and $(1, 5)$ put more probability near 1 and 0, respectively.

In R, `dbeta` evaluates the PDF and `pbeta` evaluates the CDF. The arguments `shape1` and `shape2` correspond to α and β:

```r
curve(dbeta(x, shape1 = 2, shape2 = 2), from = 0, to = 1,
      xlab = "p", ylab = "Density")
curve(pbeta(x, shape1 = 2, shape2 = 2), from = 0, to = 1,
      xlab = "p", ylab = "Probability")
```

3.9.6 The Gamma distribution

The Gamma family describes positive values with a shape parameter $a > 0$ and a rate $\lambda > 0$.

Definition 3.16: Gamma distribution

Using the Gamma function defined above, we write $T \sim \text{Gamma}(a, \lambda)$ when its PDF is

$$f_T(t) = \frac{\lambda^a}{\Gamma(a)} t^{a-1} e^{-\lambda t} \mathbf{1}_{(0,+\infty)}(t).$$

The substitution $u = \lambda t$ shows that its total integral is $\Gamma(a)/\Gamma(a) = 1$. The exponential is also a special case of the Gamma family of distributions on positive values, with shape parameter 1: since $\Gamma(1) = 1$, choosing $a = 1$ gives $\lambda e^{-\lambda t}$ for $t > 0$.

4

Random vectors

We now study several random variables observed together, such as exam result and class attendance. We want to describe their individual distributions and their dependence.

4.1 Random vectors and the plan of study

Definition 4.1: Random vector

On a common probability space $(\Omega, \mathcal{F}, P)$, a **random vector** is a vector of random variables

$$\underline{X} : \Omega \longrightarrow \mathbb{R}^p, \qquad \underline{X}(\omega) = \begin{pmatrix} X_1(\omega) \\ \vdots \\ X_p(\omega) \end{pmatrix}.$$

The integer p is the dimension of the vector. Each component $X_j : \Omega \to \mathbb{R}$ is a random variable.

There are three related tasks:

1. Describe the **joint distribution** of the whole vector.
2. Describe the distribution of each component separately: **univariate analysis**.
3. Study how the components interact or depend on one another: **multivariate analysis**.

We start with **fully discrete vectors** and $p = 2$: define the joint PMF, then derive marginal and conditional PMFs. The later cases are continuous vectors (using a joint PDF when one exists) and mixed vectors, with discrete and continuous components. Continuous marginals alone need not give a joint PDF.

Definition 4.2: Independence of random variables

Two random variables X and Y are independent when

$$P(X \in I,\ Y \in J) = P(X \in I)P(Y \in J) \quad \text{for every pair of intervals } I, J.$$

This is the independence of the events defined by the two variables. It is enough to check intervals; their probabilities determine the joint distribution. The PMF, CDF and PDF criteria below express this same condition in different ways.

4.2 Fully discrete random vectors

4.2.1 The joint PMF

Let $\underline{Z} = (X, Y)^\mathsf{T}$, where X and Y are discrete, with supports S_X and S_Y.

Definition 4.3: Joint probability mass function

The **joint PMF** assigns a probability to each pair of values:

$$p_{X,Y}(x, y) = P(X = x, Y = y) = P\big(\{X = x\} \cap \{Y = y\}\big).$$

It satisfies

$$p_{X,Y}(x, y) \geq 0, \qquad \sum_{x \in S_X} \sum_{y \in S_Y} p_{X,Y}(x, y) = 1.$$

Some pairs in $S_X \times S_Y$ may have probability zero. The joint PMF completely determines the distribution of (X, Y).

Example 21: Exam result and class attendance

Let $X = 1$ mean that a student passes the exam and $X = 0$ mean that the student fails. Let Y record class attendance: Always (A), Never (N), or Sometimes (S). These categories may be coded as 0, 1, and 2, respectively; the numerical labels are just codes.

The joint PMF is given by the following **contingency table**. Each interior cell is the probability of the corresponding pair.

	A	N	S	Total
Pass ($X = 1$)	0.09	0.20	0.01	0.30
Fail ($X = 0$)	0.05	0.60	0.05	0.70
Total	0.14	0.80	0.06	1.00

For example, $P(X = 1, Y = \text{A}) = 0.09$: the probability that a student both passes and always attends is 0.09. All six interior cells together sum to 1.

4.2.2 Marginal and conditional PMFs

4.2.2.1 Marginalization

To study X alone, we add the probabilities of all pairs with that value of X, allowing Y to take any of its possible values.

Proposition 4.1: Marginal PMFs

The marginal distributions are obtained by summing out the other variable:

$$p_X(x) = \sum_{y \in S_Y} p_{X,Y}(x, y), \qquad p_Y(y) = \sum_{x \in S_X} p_{X,Y}(x, y).$$

For a fixed x, the events $\{X = x, Y = y\}$ are disjoint as y varies, and their union is $\{X = x\}$. Hence

$$P(X = x) = \sum_{y \in S_Y} P(X = x, Y = y).$$

In the table, marginalization means taking row or column totals.

For the exam example,

$$p_X(1) = 0.09 + 0.20 + 0.01 = 0.30,$$
$$p_X(0) = 0.05 + 0.60 + 0.05 = 0.70.$$

Thus $X \sim \text{Ber}(0.30)$, with passing as the success outcome.

For attendance,

$$p_Y(\text{A}) = 0.09 + 0.05 = 0.14,$$
$$p_Y(\text{N}) = 0.20 + 0.60 = 0.80,$$
$$p_Y(\text{S}) = 0.01 + 0.05 = 0.06.$$

The marginal PMFs describe the individual variables, but they do not retain all the information in the joint table.

4.2.2.2 Conditioning

Suppose attendance is known. We then describe exam results *within that attendance group*. The conditioning value is fixed, and probabilities are renormalized within that group.

Definition 4.4: Conditional PMFs

For a value y with $p_Y(y) > 0$,

$$p_{X|Y}(x \mid y) = P(X = x \mid Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

Similarly, for x with $p_X(x) > 0$,

$$p_{Y|X}(y \mid x) = P(Y = y \mid X = x) = \frac{p_{X,Y}(x, y)}{p_X(x)}.$$

For each fixed conditioning value, the conditional probabilities are nonnegative and sum to 1 over the remaining variable.

Among students who always attend,

$$P(X = 1 \mid Y = \mathrm{A}) = \frac{0.09}{0.14} = \frac{9}{14} \approx 0.643,$$
$$P(X = 0 \mid Y = \mathrm{A}) = \frac{0.05}{0.14} = \frac{5}{14} \approx 0.357.$$

Among students who never attend,

$$P(X = 1 \mid Y = \mathrm{N}) = \frac{0.20}{0.80} = 0.25, \qquad P(X = 0 \mid Y = \mathrm{N}) = \frac{0.60}{0.80} = 0.75.$$

The same calculation for the remaining column gives

$$P(X = 1 \mid Y = \mathrm{S}) = \frac{0.01}{0.06} = \frac{1}{6}, \qquad P(X = 0 \mid Y = \mathrm{S}) = \frac{0.05}{0.06} = \frac{5}{6}.$$

Thus each conditional distribution is Bernoulli:

$$X \mid (Y = y) \sim \mathrm{Ber}\big(\pi(y)\big), \qquad \pi(y) = P(X = 1 \mid Y = y).$$

Its success probability depends on the attendance group. In particular, 0.643, 0.25, and 1/6 differ from the marginal success probability 0.30.

4.2.3 Independence and the role of dependence

Definition 4.5: Independence of discrete random variables

Two discrete random variables X and Y are independent if and only if

$$p_{X,Y}(x,y) = p_X(x)p_Y(y) \qquad \text{for every } (x,y) \in S_X \times S_Y.$$

Equivalently,

$$p_{X|Y}(x \mid y) = p_X(x) \quad \text{for every } x \text{ and every } y \text{ with } p_Y(y) > 0.$$

Under independence, knowing one variable does not change the distribution of the other. The factorization must hold for every pair of values.

The original exam table is not independent, because

$$p_{X,Y}(1, \mathrm{A}) = 0.09 \neq 0.30 \cdot 0.14 = 0.042.$$

An independent table with the same marginals.

Keep the same row and column totals, and define a new joint PMF by multiplying the marginals:

$$\widetilde{p}_{X,Y}(x,y) = p_X(x)p_Y(y).$$

This gives

	A	N	S	Total
Pass ($X = 1$)	0.042	0.240	0.018	0.300
Fail ($X = 0$)	0.098	0.560	0.042	0.700
Total	0.140	0.800	0.060	1.000

For instance, $\widetilde{p}_{X,Y}(0, \mathrm{S}) = 0.70 \cdot 0.06 = 0.042$. Every conditional passing probability in this new table is 0.30.

How far are we from independence?

The two tables have identical marginals but different joint distributions. This shows that the marginals alone do not determine the dependence between the variables.

One way to study departures from independence is to compare the actual joint PMF with the independent one:

$$D\big(p_{X,Y}, \widetilde{p}_{X,Y}\big), \qquad \widetilde{p}_{X,Y} = p_X p_Y.$$

Here D denotes a chosen discrepancy between distributions. Different choices emphasize different departures from independence.

To describe dependence beyond a discrete table, we first introduce the joint CDF, which applies to every pair of random variables. After the continuous and mixed cases, copulas will describe how marginals combine with a dependence structure.

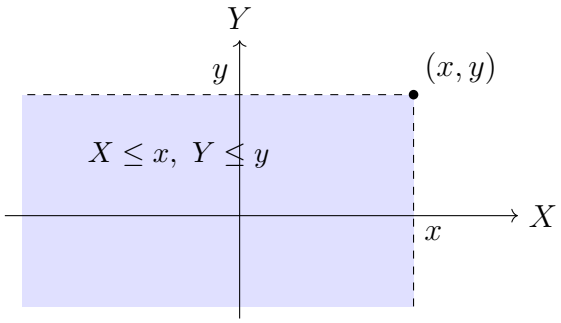
4.3 The joint CDF

Definition 4.6: Joint cumulative distribution function

For any pair of random variables, discrete or otherwise,

$$F_{X,Y}(x,y) = P(X \le x, Y \le y), \qquad (x,y) \in \mathbb{R}^2.$$

This is the probability of the lower-left region $(-\infty, x] \times (-\infty, y]$ in the plane.



4.3.1 Recovering the marginals

To study X alone, we remove the restriction on Y by letting its threshold tend to $+\infty$.

Proposition 4.2: Marginal CDFs

The marginal CDFs are obtained as limits of the joint CDF:

$$F_X(x) = \lim_{y \to +\infty} F_{X,Y}(x,y), \qquad F_Y(y) = \lim_{x \to +\infty} F_{X,Y}(x,y).$$

As y increases, the event $\{X \le x,\ Y \le y\}$ grows and approaches $\{X \le x\}$, so its probability approaches $P(X \le x) = F_X(x)$.

4.3.2 Basic properties

The joint CDF takes values in $[0,1]$, is nondecreasing in each coordinate, and is right-continuous. It tends to 0 when either threshold tends to $-\infty$, and to 1 when both tend to $+\infty$.

In two dimensions, probabilities of rectangles must also be nonnegative. For $a < b$ and $c < d$,

$$\begin{aligned} P(a < X \le b,\ c < Y \le d) &= F_{X,Y}(b,d) - F_{X,Y}(a,d) \\ &\quad - F_{X,Y}(b,c) + F_{X,Y}(a,c) \ge 0. \end{aligned}$$

This is called the **2-increasing property**; being nondecreasing separately in each coordinate is not sufficient.

Proposition 4.3: X, Y independent $\iff$ joint CDF $= F_X F_Y$

Let X and Y be **arbitrary random variables**, with CDFs F_X and F_Y. They are **independent if and only if** their joint CDF satisfies

$$F_{X,Y}(x, y) = F_X(x)F_Y(y) \qquad \text{for all } x, y \in \mathbb{R}.$$

Proof. If X and Y are independent, the definition applied to the events $X \le x$ and $Y \le y$ gives

$$F_{X,Y}(x, y) = P(X \le x, Y \le y) = P(X \le x)P(Y \le y) = F_X(x)F_Y(y).$$

Conversely, substitute this product into the rectangle formula above. It gives

$$\begin{aligned} P(a < X \le b,\ c < Y \le d) &= [F_X(b) - F_X(a)][F_Y(d) - F_Y(c)] \\ &= P(a < X \le b)P(c < Y \le d). \end{aligned}$$

Thus the interval probabilities factorize, which is independence. Limits include unbounded intervals, and differences of limits include their endpoint variants. □

4.4 Fully continuous random vectors

4.4.1 Joint probability density functions

We now study continuous components of the random vectors introduced above. We have already studied joint, marginal and conditional PMFs for discrete variables. The next sections ask the same questions for continuous variables: how do we describe them together, recover one component, and use information about another component? Probabilities will be computed by integrating densities, rather than summing masses.

Let $\underline{X} = (X_1, \dots, X_p)^\mathsf{T}$ be a random vector with p components. A possible value is $\underline{x} = (x_1, \dots, x_p)^\mathsf{T} \in \mathbb{R}^p$. Here we assume that its joint distribution admits a PDF. For $p = 2$, we write the components as (X, Y) and the density as $f_{X,Y}(x, y)$.

Definition 4.7: Joint PDF

A **joint probability density function** $f_{\underline{X}} : \mathbb{R}^p \to [0, +\infty)$ satisfies

$$\int_{\mathbb{R}^p} f_{\underline{X}}(\underline{x})\, d\underline{x} = 1,$$

and gives the probability of a measurable region $A \subseteq \mathbb{R}^p$ through

$$P(\underline{X} \in A) = \int_A f_{\underline{X}}(\underline{x})\, d\underline{x}.$$

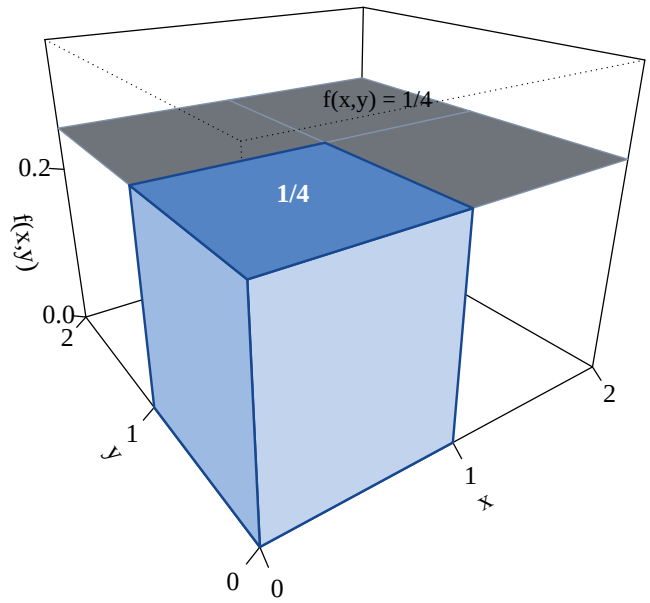
In two dimensions, $d\underline{x}$ means $dx\, dy$. A joint PDF assigns density to locations in the plane or in a higher-dimensional space. A probability is obtained by integrating over a region.

In particular, a small rectangle near (x, y) has probability approximately $f_{X,Y}(x, y)\, \Delta x\, \Delta y$ when the density varies little over the rectangle. The PDF value itself is not a point probability; in this setting, every individual point has probability zero.

Example 22

Choose a point uniformly at random in the square $[0, 2] \times [0, 2]$. Let X be its horizontal coordinate and Y its vertical coordinate. Together they form the random vector $\underline{X} = (X, Y)$. Regions with equal area have equal probability. The square has area $2 \cdot 2 = 4$, so the constant density must be $1/4$:

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{4}, & (x, y) \in [0, 2]^2, \\ 0, & \text{otherwise.} \end{cases}$$



Step 1: read the density as a height. The horizontal axes of the graph show x and y. The vertical axis shows $f_{X,Y}(x, y)$: the density is the height of the surface. Here that height is $1/4$ throughout the square. We have two random variables; the third axis displays their density.

Step 2: check the total probability. The probability of a region is the volume under the surface, above that region. Since the height is constant, we multiply the height by the area of the base. Over the whole square,

$$\int_0^2 \int_0^2 \frac{1}{4}\, dy\, dx = \underbrace{\frac{1}{4}}_{\text{height}} \cdot \underbrace{(2 \cdot 2)}_{\text{area of the base}} = 1.$$

Here $dx\, dy$ is an element of area in the plane of the two coordinates.

Step 3: find the probability of a smaller region. What is the probability that both coordinates lie between 0 and 1? The base is now the smaller square $[0, 1] \times [0, 1]$, with area 1. The blue block in the graph has this base and height $1/4$. Its volume gives the probability. Since X and Y are nonnegative, this probability is $F_{X,Y}(1, 1)$. Using the definition of the joint CDF,

$$F_{X,Y}(1,1) = \int_{-\infty}^{1} \int_{-\infty}^{1} f_{X,Y}(x, y)\, dy\, dx = \int_0^1 \int_0^1 \frac{1}{4}\, dy\, dx = \int_0^1 \frac{1}{4}\, dx = \boxed{\frac{1}{4}}.$$

The lower limits become 0 because the density is zero outside the square $[0, 2]^2$. If the density varies across a region, the surface has different heights. The integral adds up the small volumes to obtain the probability.

Note: The joint density assumption

Having continuous marginals alone does not guarantee a joint PDF.

Choose X uniformly between 0 and 2, then set $Y = X$. Both variables are continuous, but the possible pairs are $(0.3, 0.3)$, $(1.2, 1.2)$, and so on: they all lie on the diagonal $y = x$. Let

$$A = \{(x, y) : 0 \le x \le 2,\ y = x\}.$$

Which points belong to A? If $x = 0.5$, only $(0.5, 0.5)$; if $x = 1$, only $(1, 1)$. So A is **just the diagonal segment**.

The segment has length, but **no thickness**, so its area is zero. We can surround it with a strip and make that strip as thin as we like: its area approaches zero while it still contains the whole segment. The word "region" here simply means a set of points; it does not imply a positive area.

By the definition of a joint PDF, the integral over **the whole plane** must equal 1:

$$\iint_{\mathbb{R}^2} f_{X,Y}(x, y)\, dx\, dy = 1.$$

In our example, the pair always lands in A, because $Y = X$. All the probability is on the diagonal, so $P\big((X, Y) \in A\big) = 1$.

Now suppose a joint PDF existed. By its definition, integrating it over A would have to give that probability. But A has area zero, so the integral would be zero. This gives the contradiction

$$1 = P\big((X, Y) \in A\big) \overset{\text{def.}}{=} \iint_A f_{X,Y}(x, y)\, dx\, dy = 0.$$

No joint PDF can satisfy this requirement. The pair has a joint distribution, but **no ordinary joint PDF**.

4.4.2 Normalizing a nonnegative function

We have already met normalization with conditional PMFs: dividing by the probability of the conditioning group makes the probabilities within that group sum to 1.

Here we want a density whose total integral is 1. Sometimes a model is supplied only up to a multiplicative constant.

Definition 4.8: Kernel of a PDF

A **kernel of a PDF** is a nonnegative function $g(\underline{x})$ proportional to that PDF. It specifies the shape of the density, leaving a positive multiplicative constant to be determined.

To obtain the PDF, compute the total integral

$$c = \int_{\mathbb{R}^p} g(\underline{x})\, d\underline{x}.$$

If $0 < c < +\infty$, then

$$f_{\underline{X}}(\underline{x}) = \frac{g(\underline{x})}{c}$$

is the corresponding PDF. The constant c is the **normalizing constant**.

The resulting function remains nonnegative, and its total integral is $c/c = 1$.

Example 23

Start with one variable and the nonnegative function

$$g(x) = \begin{cases} x, & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

This is the **kernel**: it gives the shape of the density, while the factor that makes its total integral 1 is still missing.

Step 1: find the total area.

$$c = \int_{\mathbb{R}} g(x)\, dx = \int_0^2 x\, dx = \left[\frac{x^2}{2}\right]_0^2 = \frac{4}{2} - 0 = 2.$$

The function is not yet a PDF: its total area is 2, rather than 1.

Step 2: divide every height by 2.

$$f_X(x) = \frac{g(x)}{2} = \begin{cases} \frac{x}{2}, & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

The function remains nonnegative, and now

$$\int_{\mathbb{R}} f_X(x)\, dx = \int_0^2 \frac{x}{2}\, dx = \frac{1}{2}\int_0^2 x\, dx = \frac{1}{2}\cdot 2 = 1.$$

So f_X is a PDF.

4.4.3 Moving between the joint CDF and the joint PDF

Recall that the joint CDF is $F_{X,Y}(x,y) = P(X \le x, Y \le y)$. With a joint PDF, this probability is the integral over the lower-left region:

$$F_{X,Y}(x,y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{X,Y}(s,t)\,dt\,ds.$$

The letters s,t are integration variables; x,y are the fixed thresholds defining the region. For p components, the same idea uses p integrals, one for each coordinate.

Where the required derivatives exist, differentiating with respect to both thresholds recovers the joint density:

$$f_{X,Y}(x,y) = \frac{\partial^2}{\partial x\,\partial y} F_{X,Y}(x,y).$$

This extends the univariate relation $f_X(x) = F_X'(x)$. In general, the density identity is understood almost everywhere; the examples below are smooth inside their supports. Here **almost everywhere** allows exceptions on a set of area zero, such as individual points or boundary lines. Changing density values only on such a set does not change any probability integral. Inside the supports of these examples, the derivatives are ordinary continuous derivatives; the boundaries can be treated separately.

4.4.4 Marginal densities

We know how X and Y behave together; how do we describe X alone (or, similarly, Y alone)?

To study X alone, we must allow Y to take every possible value. For discrete variables we summed over those possibilities; for a joint PDF we integrate over them.

Proposition 4.4: Marginal PDFs

The marginal densities are obtained by integrating out the other variable:

$$f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x,y)\,dy, \qquad f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x,y)\,dx.$$

In the first integral, x stays fixed while y is integrated out. The integration can be restricted to the support because the joint density is zero elsewhere. Both marginal densities are normalized: integrating a marginal over its remaining coordinate gives the total joint integral, which is 1.

The procedure also works when we retain several components. For a vector (X_1, X_2, X_3) with joint PDF $f_{1,2,3}$, the pair (X_1, X_2) has density

$$f_{1,2}(x_1,x_2) = \int_{\mathbb{R}} f_{1,2,3}(x_1,x_2,t)\,dt.$$

We integrate out the component we do not want to retain.

4.4.5 Conditional densities

We now ask how the distribution of X changes when the value of Y is known. In the discrete case we divided a joint probability by $P(Y = y)$. For a continuous Y, however, $P(Y = y) = 0$; substituting point probabilities into that formula would not define the desired conditional distribution.

For a fixed threshold x, first condition on the interval $y < Y \le y + h$, which has positive probability when its density is positive nearby. Then

$$P(X \le x \mid y < Y \le y + h) = \frac{\displaystyle\int_{y}^{y+h} \int_{-\infty}^{x} f_{X,Y}(s,t)\, ds\, dt}{\displaystyle\int_{y}^{y+h} f_Y(t)\, dt}.$$

At values y where these averages converge and $f_Y(y) > 0$, letting $h \to 0^+$ gives the conditional CDF

$$F_{X|Y}(x \mid y) = \frac{\displaystyle\int_{-\infty}^{x} f_{X,Y}(s,y)\, ds}{f_Y(y)}.$$

Differentiating in x gives the density ratio below. This explains the passage from a small interval to the notation "given $Y = y$".

The density formula can be understood by conditioning on a small interval near y. Its probability is approximately $f_Y(y)\,\Delta y$, while the probability of a small joint rectangle near (x, y) is approximately $f_{X,Y}(x, y)\,\Delta x\,\Delta y$. Their ratio is

$$\frac{f_{X,Y}(x,y)\,\Delta x\,\Delta y}{f_Y(y)\,\Delta y} = \frac{f_{X,Y}(x,y)}{f_Y(y)}\,\Delta x.$$

The coefficient of Δx is the conditional density.

Definition 4.9: Conditional PDFs

When a joint PDF exists, a conditional density of X given $Y = y$ can be defined, for $f_Y(y) > 0$, by

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}.$$

Similarly, for $f_X(x) > 0$,

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)}.$$

The conditioning value is held fixed. Integrating the conditional density over the other variable gives 1.

For instance, the normalization follows from marginalization:

$$\int_{\mathbb{R}} f_{X|Y}(x \mid y)\, dx = \frac{\int_{\mathbb{R}} f_{X,Y}(x,y)\, dx}{f_Y(y)} = 1.$$

4.4.6 Independence for variables with PDFs

Independence means that learning where X lies does not change the probabilities for Y.

Proposition 4.5: X, Y independent $\iff$ joint PDF $= f_X f_Y$

Suppose X and Y **each have a PDF**, denoted by f_X and f_Y. They are **independent if and only if** their joint PDF exists and satisfies

$$f_{X,Y}(x,y) = f_X(x) f_Y(y) \qquad \text{almost everywhere.}$$

Proof. ($\Rightarrow$) **Independence implies factorization.**

Take an interval I for X and an interval J for Y. The event $X \in I$, $Y \in J$ means that the pair falls in the rectangle $I \times J$. Independence gives

$$\begin{aligned} P(X \in I,\ Y \in J) &= P(X \in I) P(Y \in J) \\ &= \left(\int_I f_X(x)\, dx \right) \left(\int_J f_Y(y)\, dy \right) \\ &= \iint_{I \times J} f_X(x) f_Y(y)\, dx\, dy. \end{aligned}$$

The product therefore gives exactly the joint probabilities we need. Since these rectangle probabilities determine the joint distribution, $f_X(x) f_Y(y)$ is a joint PDF for the pair.

($\Leftarrow$) **Factorization implies independence.**

Now suppose the joint PDF exists and equals $f_X(x) f_Y(y)$ almost everywhere. For any intervals I and J,

$$\begin{aligned} P(X \in I,\ Y \in J) &= \int_I \int_J f_{X,Y}(x,y)\, dy\, dx \\ &= \int_I \int_J f_X(x) f_Y(y)\, dy\, dx \\ &= \left(\int_I f_X(x)\, dx \right) \left(\int_J f_Y(y)\, dy \right) \\ &= P(X \in I) P(Y \in J). \end{aligned}$$

The integrals separate because f_X depends only on x and f_Y only on y. Since the last equality holds for every pair of intervals, X and Y are independent. □

4.4.7 Two independent exponential variables

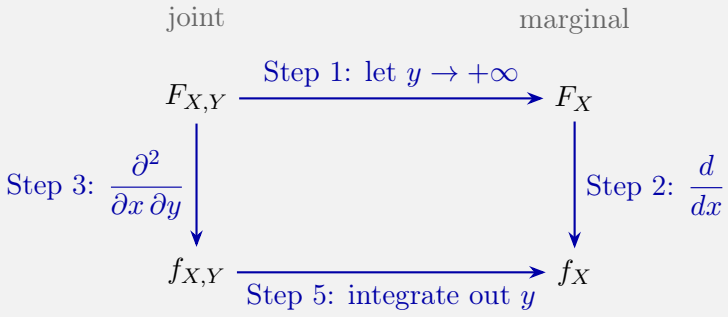
Example 24: Two independent exponential variables

Consider the following joint CDF, with parameters $\alpha > 0$ and $\beta > 0$:

$$F_{X,Y}(x,y) = \begin{cases} 1 - e^{-\alpha x} - e^{-\beta y} + e^{-(\alpha x + \beta y)}, & x > 0,\ y > 0, \\ 0, & x \le 0 \text{ or } y \le 0. \end{cases}$$

Find the marginal CDFs and PDFs, recover the joint PDF, and check whether X and Y are independent.

The plan. Four functions describe X in this example: the joint and the marginal CDF in the top row, the joint and the marginal PDF in the bottom row. We start from $F_{X,Y}$ and follow the arrows.



Two paths lead from $F_{X,Y}$ to f_X.

Step 1: marginal CDFs. By the marginal CDF proposition, we remove the restriction on Y by letting $y \to +\infty$. Both terms containing $e^{-\beta y}$ vanish:

$$F_X(x) = \lim_{y \to +\infty} F_{X,Y}(x,y) = 1 - e^{-\alpha x}, \qquad x > 0.$$

Letting $x \to +\infty$ instead gives

$$F_Y(y) = \lim_{x \to +\infty} F_{X,Y}(x,y) = 1 - e^{-\beta y}, \qquad y > 0.$$

Both marginal CDFs are zero when their argument is ≤ 0.

Step 2: marginal PDFs. Differentiating each marginal CDF gives

$$f_X(x) = \alpha e^{-\alpha x} \mathbf{1}_{(0,+\infty)}(x), \qquad f_Y(y) = \beta e^{-\beta y} \mathbf{1}_{(0,+\infty)}(y).$$

These are **exponential distributions** with rates α and β: $X \sim \text{Exp}(\alpha)$ and $Y \sim \text{Exp}(\beta)$.

Step 3: joint PDF. Differentiate the joint CDF first in x, then in y. Inside the positive quadrant,

$$\frac{\partial F_{X,Y}}{\partial x}(x,y) = \alpha e^{-\alpha x} - \alpha e^{-(\alpha x + \beta y)},$$

$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}}{\partial x\,\partial y}(x,y) = \alpha\beta e^{-(\alpha x + \beta y)}.$$

In the derivative in y, the term $\alpha e^{-\alpha x}$ does not depend on y and vanishes. The joint density is zero outside the quadrant.

Step 4: independence. Both the joint CDF and the joint PDF factorize.

Through the CDF. Write $a = e^{-\alpha x}$ and $b = e^{-\beta y}$, so that $e^{-(\alpha x+\beta y)} = ab$. For $x, y > 0$,

$$\begin{aligned} F_{X,Y}(x,y) &= 1 - a - b + ab = (1-a) - b(1-a) = (1-a)(1-b) \\ &= \big(1 - e^{-\alpha x}\big)\big(1 - e^{-\beta y}\big) = F_X(x) F_Y(y). \end{aligned}$$

When $x \le 0$ or $y \le 0$, both sides are zero, so the factorization holds for all x, y. By the CDF independence criterion proved above, X and Y are therefore independent.

Through the PDF. The exponential of a sum is a product of exponentials:

$$f_{X,Y}(x,y) = \big[\alpha e^{-\alpha x}\mathbf{1}_{(0,+\infty)}(x)\big]\big[\beta e^{-\beta y}\mathbf{1}_{(0,+\infty)}(y)\big] = f_X(x) f_Y(y).$$

By the PDF independence criterion proved above, X and Y are therefore independent.

Through the conditional densities. Independence also shows up in the conditional densities. For $y > 0$,

$$f_{X|Y}(x \mid y) = \frac{\alpha\beta e^{-(\alpha x+\beta y)}}{\beta e^{-\beta y}}\mathbf{1}_{(0,+\infty)}(x) = f_X(x).$$

Knowing Y leaves the distribution of X unchanged. The same holds for $f_{Y|X}(y \mid x) = f_Y(y)$.

Step 5: check by marginalizing. Steps 1 and 2 reached f_X through the marginal CDF. We can also reach it from the joint PDF of Step 3: by the marginal PDF proposition, we integrate out y. The density is zero for $y \le 0$, so the integral runs over $y > 0$:

$$f_X(x) = \int_0^{+\infty} \alpha\beta e^{-(\alpha x+\beta y)}\, dy = \alpha e^{-\alpha x} \underbrace{\int_0^{+\infty} \beta e^{-\beta y}\, dy}_{=1} = \alpha e^{-\alpha x}, \qquad x > 0.$$

The inner integral is 1 because $\beta e^{-\beta y}$ is the density of Y. Both paths give the same f_X, so the computations agree.

4.5 Mixed random vectors

4.5.1 A binomial model with a latent probability

We now study the mixed vector $\underline{Z} = (X, P)^\mathsf{T}$. Here X is an observable, discrete count, while P is a continuous, **latent** variable: it is not directly observed, but represents uncertainty about the success probability underlying the count. The capital letter P

denotes a random variable, and p denotes a possible value of it. We use $\Pr(\cdot)$ for the probability operator in this example to distinguish it from the variable P.

Example 25: A coin of unknown bias

We have a coin and do not know how biased it is. Chance decides two things, one after the other:

1. first, **how biased the coin is**: this is P, the probability of heads, a **real** number in the interval $(0, 1)$, such as 0.3 or 0.37, so a **continuous** variable;
2. then, **how many heads** appear when the coin is tossed n times: this is X, a count, so a **discrete** variable.

The pair (X, P) is called mixed because one coordinate is discrete and the other is continuous.

4.5.1.1 The conditional model for the observable

Fix a positive integer n, the number of trials. Conditional on $P = p$, each trial is a success with probability p and a failure with probability $1 - p$, and the trials are independent. Let X count the successes. Then

$$X \mid (P = p) \sim \mathrm{Bin}(n, p).$$

This is the binomial distribution introduced in Section 3.9.4. Write its conditional PMF as

$$b(k \mid p) = \Pr(X = k \mid P = p) = \binom{n}{k} p^k (1 - p)^{n-k}, \qquad k = 0, \ldots, n, \quad 0 < p < 1.$$

4.5.1.2 The distribution of the latent variable

We also need a PDF $\pi(p)$ for P, called its **prior density**: it describes our uncertainty before observing X. Consider two choices:

$$P \sim \mathrm{Unif}(0, 1) \qquad \text{or} \qquad P \sim \mathrm{Beta}(\alpha, \beta), \quad \alpha, \beta > 0.$$

For the uniform choice, $\pi(p) = 1$ on $(0, 1)$ and zero elsewhere. The Beta family introduced in Section 3.9.5 allows different shapes on the same interval.

4.5.1.3 Combine the PMF and the PDF

Because X is discrete and P is continuous, their joint model uses a mass in the first coordinate and a density in the second.

Definition 4.10: Mixed joint function

Let $X \mid (P = p)$ have PMF $b(k \mid p)$ on $\{0, \dots, n\}$, and let P have prior PDF $\pi(p)$ on $(0, 1)$. The **mixed joint function** of (X, P) is

$$q(k, p) = b(k \mid p)\, \pi(p), \qquad k = 0, \dots, n, \quad 0 < p < 1.$$

It gives the probability of a count and an interval:

$$\Pr(X = k,\ a < P < b) = \int_a^b q(k, p)\, dp, \qquad 0 \le a < b \le 1.$$

Note: Conditional × marginal

The structure in Definition 4.10 is not new: a joint distribution is a conditional times a marginal. Rearranging Definition 4.4 of conditional PMFs and Definition 4.9 of conditional PDFs gives

$$p_{X,Y}(x, y) = p_{X|Y}(x \mid y)\, p_Y(y), \qquad f_{X,Y}(x, y) = f_{X|Y}(x \mid y)\, f_Y(y).$$

Here the conditional is the PMF $b(k \mid p)$ and the marginal is the PDF $\pi(p)$: first the latent variable, then the count given its value.

This function is not an ordinary two-dimensional PDF: probabilities require a *sum* over the count and an *integral* over the latent probability. It is normalized because

$$\sum_{k=0}^{n} \int_0^1 q(k, p)\, dp = \int_0^1 \pi(p) \underbrace{\sum_{k=0}^{n} b(k \mid p)}_{=1}\, dp = 1.$$

4.5.1.4 Marginalization and conditioning in the mixed model

To describe X alone, average over the unknown success probability. Its marginal PMF, denoted by $m(k)$, is

$$m(k) = \Pr(X = k) = \int_0^1 b(k \mid p)\, \pi(p)\, dp.$$

This is the continuous version of summing over an unobserved component. Conversely, summing $q(k, p)$ over k recovers $\pi(p)$.

After observing $X = k$, we want to update the distribution of P. For $m(k) > 0$, its **posterior density** is

$$f_{P|X}(p \mid k) = \frac{b(k \mid p)\, \pi(p)}{m(k)}.$$

The count k is now fixed, while p varies. The numerator gives more weight to values of p that make the observed count likely; the denominator normalizes those weights to integrate to 1. This is Bayes' rule for a discrete observation and a continuous latent variable.

The Beta family is especially useful in this binomial model: after observing a count, its posterior also belongs to the Beta family. This property is called **conjugacy**; identifying the updated parameters is one of the final exercise questions.

Equivalently, the same joint function can be written in the other order:

$$q(k, p) = f_{P|X}(p \mid k)\, m(k).$$

The first factorization uses a conditional PMF and a marginal PDF; the second uses a conditional PDF and a marginal PMF. Both describe the same joint distribution. A conditional factorization by itself does not assert independence.

Questions left to work through.

To complete the example, determine:

1. What is the marginal distribution of X?
2. What is the conditional distribution of $P \mid X$?

Apply the formulas above to each proposed prior, keeping track of which object is a PMF and which is a PDF.

4.6 Copulas and Sklar's theorem

The marginals describe the individual variables. A copula provides a way to combine those marginals into a joint CDF, accounting for their dependence.

Definition 4.11: Bivariate copula

A **copula** is a joint CDF of two random variables with standard uniform marginals:

$$C(u, v) = P(U \le u, V \le v), \qquad U, V \sim \text{Unif}(0, 1).$$

The variables U and V need not be independent. On $[0, 1]^2$, the marginal conditions are $C(u, 1) = u$ and $C(1, v) = v$.

Theorem 4.1: Sklar's theorem: bivariate form

For any joint CDF $F_{X,Y}$ with marginals F_X and F_Y, there exists a copula C such that

$$F_{X,Y}(x, y) = C\big(F_X(x), F_Y(y)\big).$$

If both marginals are continuous, the copula is unique. Otherwise, it is uniquely determined only on $\text{Ran}(F_X) \times \text{Ran}(F_Y)$, the product of the sets of values taken by the marginal CDFs.

Conversely, combining any two marginal CDFs with a copula using this formula produces a joint CDF with those marginals.

The schematic statement

$$\text{joint distribution} = \text{marginals} + \text{dependence structure}$$

is made precise by this composition. The plus sign in the schematic statement does not mean addition of CDFs.

For independence, the **independence copula** is

$$C_{\perp}(u, v) = uv.$$

Substituting the marginals gives

$$F_{X,Y}(x, y) = C_{\perp}\big(F_X(x), F_Y(y)\big) = F_X(x)F_Y(y).$$

This is the CDF counterpart of the independent PMF table.

4.7 IID samples from a population

We now connect population distributions to the data used in statistical learning. Let F_X be the CDF of a population variable X. A sampling scheme produces observations from that population.

Definition 4.12: Independent and identically distributed sample

A random sample $(X_1, \ldots, X_n)$ is **IID** from F_X if

- the random variables $X_1, \ldots, X_n$ are mutually independent;
- each X_i has the same CDF F_X.

Here n is the sample size. Uppercase X_i denotes a random observation before sampling; lowercase x_i denotes its observed value. The realized dataset is $\mathcal{D}_n = (x_1, \ldots, x_n)$.

For n variables, **mutual independence** means that

$$P\left(\bigcap_{i=1}^{n}\{X_i \in I_i\}\right) = \prod_{i=1}^{n} P(X_i \in I_i) \quad \text{for all intervals } I_1, \ldots, I_n.$$

Taking any unused interval to be $\mathbb{R}$ also gives the factorization for every subcollection.

4.7.1 The joint CDF of the sample

Independence gives a product, and identical distribution gives the same CDF in every factor:

$$\begin{aligned} F_{X_1,\ldots,X_n}(x_1, \ldots, x_n) &= P(X_1 \le x_1, \ldots, X_n \le x_n) \\ &= \prod_{i=1}^{n} P(X_i \le x_i) \\ &= \prod_{i=1}^{n} F_X(x_i). \end{aligned}$$

This is the probability that *all* observations lie below their respective thresholds.

4.7.2 Multivariate observations

Each observation may itself be a vector $\underline{X}_i \in \mathbb{R}^p$. If these vectors are IID with common joint CDF $F_{\underline{X}}$, then

$$P(\underline{X}_1 \le \underline{x}_1, \ldots, \underline{X}_n \le \underline{x}_n) = \prod_{i=1}^{n} F_{\underline{X}}(\underline{x}_i),$$

where each vector inequality is interpreted componentwise. The product is over observations. Coordinates within one observation may still be dependent.

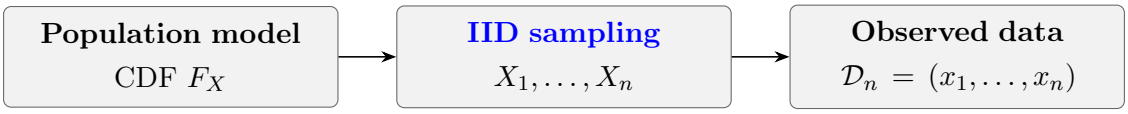
Thus n counts observations, whereas p counts variables in each observation.

4.8 Likelihood and the empirical distribution

Our goal is to **estimate the population distribution from the data we have observed**. We have collected measurements and want to use them to learn about the process that produced them.

For example, let X be the waiting time until an event occurs. We collect n waiting times, called $x_1, \ldots, x_n$. We want to build a model that lets us answer questions such as: *"What is the probability of waiting less than five minutes?"* The observed times are our data; the distribution of waiting times in the population is what we want to estimate.

If the population variable has a PDF, we denote it by f_X; its CDF is F_X. These describe the same distribution in different ways. The PDF describes how probability is concentrated, while the CDF gives the probability of being at or below a threshold. Both describe the population, beyond the particular observations we have collected.



The diagram shows how a population model leads to observations. Here n is the sample size. Before sampling, the X_i are independent random variables sharing the population CDF F_X; after sampling, the x_i are the observed values. The population model and the sampling scheme together describe the **data-generating process**: what varies in the population and how we obtain observations from it.

Learning goes in the other direction: we start with the observed data on the right and try to estimate the population distribution on the left. For the waiting-time example, we use the recorded times to estimate f_X, then use the fitted model to calculate probabilities for waiting times.

This is the **learning problem**. The observations are fixed; we compare possible distributions to find a model that fits them.

One rule for choosing among these candidates is **maximum likelihood**: choose the model that makes the observed data most plausible according to their likelihood. We introduce this comparison formally below, distinguishing masses, densities and CDF values.

Note: Joint CDF and likelihood

The product $\prod_i F_X(x_i)$ above is the joint CDF evaluated at thresholds. For exact observations, the usual IID likelihood is instead a product of **PMF or PDF values**. The two objects must be distinguished when reading the handwritten notes.

4.8.1 Maximum likelihood estimation

Suppose our candidates form a family of distributions described by a parameter θ. This parameter may be one number or several numbers; Θ is the set of values we allow it to

take. Choosing a value of θ chooses one candidate distribution. For one observation, write p_θ for its PMF in a discrete model, or f_θ for its PDF in a continuous model.

For each candidate, we evaluate its PMF or PDF at all the observed values. The IID assumption makes their joint mass or density a product. This product is the candidate's likelihood: the score we use to compare it with the other candidates.

Definition 4.13: Likelihood

For fixed observed data $\mathcal{D}_n = (x_1, \ldots, x_n)$ and a model parameter $\theta \in \Theta$, the **IID likelihood** is

$$L(\theta; x_1, \ldots, x_n) = \begin{cases} \displaystyle\prod_{i=1}^{n} p_\theta(x_i), & \text{discrete model,} \\ \displaystyle\prod_{i=1}^{n} f_\theta(x_i), & \text{model with a PDF.} \end{cases}$$

Definition 4.14: Maximum likelihood estimation

A **maximum likelihood estimator (MLE)** chooses

$$\widehat{\theta}_n \in \arg\max_{\theta \in \Theta} L(\theta; x_1, \ldots, x_n),$$

when a maximizer exists. Here $\widehat{\theta}_n$ is the parameter estimate calculated from the n observations, and arg max denotes the set of parameter values attaining the largest likelihood.

When a candidate is denoted directly by its PDF $f = f_\theta$, we write $L(f; \mathcal{D}_n)$ for the same likelihood and $\widehat{f}$ for the fitted density.

The data stay fixed while we vary the model parameter. For our waiting times, this means evaluating the same recorded times under different candidate distributions, then selecting a candidate with the largest likelihood. In the continuous case, a PDF value is a density, not the probability of that exact observed value. Equivalently, one may maximize the log-likelihood: the logarithm turns products into sums while preserving the ordering of positive likelihoods.

Example 26: MLE from two waiting times

We observe two independent waiting times:

$$x_1 = 1 \text{ minute}, \qquad x_2 = 2 \text{ minutes}.$$

Assume they are IID from a uniform distribution on $[0, \theta]$. Here $\theta > 0$ is unknown: it is the largest waiting time allowed by our model. Changing θ changes the candidate distribution.

Which value of θ does maximum likelihood choose?

Compare the same data under different candidates.

Step 1: Write the density of one observation. A uniform PDF is constant on its interval. Its area is 1, so its height is $1/\theta$:

$$f_\theta(x) = \frac{1}{\theta}\mathbf{1}_{[0,\theta]}(x).$$

The parameter θ is the candidate interval's upper limit; x is a possible waiting time.

Step 2: Check which candidates can include our observations. If $\theta < 2$, the second observed time lies outside the interval, so $f_\theta(2) = 0$. If $\theta \geq 2$, both observations are included and each has density $1/\theta$.

Step 3: Calculate the likelihood. Independence gives a product:

$$L(\theta; 1, 2) = f_\theta(1) f_\theta(2) = \begin{cases} 0, & 0 < \theta < 2, \\ 1/\theta^2, & \theta \geq 2. \end{cases}$$

For example, the same data give the following scores:

Candidate θ	Candidate interval	Likelihood
1	$[0, 1]$	0
2	$[0, 2]$	$1/4 = 0.25$
3	$[0, 3]$	$1/9 \approx 0.111$
10	$[0, 10]$	$1/100 = 0.01$

These scores are products of density values, not probabilities of the exact observations.

Step 4: Choose the parameter with the highest score. Among all $\theta \geq 2$, increasing θ makes $1/\theta^2$ smaller. The best candidate is therefore the smallest upper limit that includes both observations:

$$\boxed{\widehat{\theta}_2 = 2 \text{ minutes}.}$$

The MLE has selected the fitted model $\text{Unif}(0, 2)$. We kept the observations 1 and 2 fixed and varied the model's parameter to find its largest likelihood. The result is an estimate under the uniform assumption; new observations could produce a different estimate of the upper limit.

Example 27: Why does likelihood prefer one density?

The earlier waiting-time example showed how to maximize likelihood over a parameter. Here we focus on why one candidate receives a higher score than another.

We observe four IID waiting times, measured in minutes:

$$\mathcal{D}_4 = (4, 12, 7, 3).$$

Consider exactly two candidate models: a uniform distribution on $[0, 15]$ and a uniform distribution on $[0, 30]$. Their PDFs are

$$f_1(x) = \frac{1}{15}\mathbf{1}_{[0,15]}(x), \qquad f_2(x) = \frac{1}{30}\mathbf{1}_{[0,30]}(x).$$

Our class is $\mathcal{F} = \{f_1, f_2\}$. Each uniform PDF is a rectangle with total area 1, so its height is the reciprocal of its interval's width.

Step 1: evaluate each candidate at the same observations. All four observations lie in both intervals. Therefore,

$$f_1(4) = f_1(12) = f_1(7) = f_1(3) = \frac{1}{15},$$

whereas

$$f_2(4) = f_2(12) = f_2(7) = f_2(3) = \frac{1}{30}.$$

The observations stay fixed; only the candidate density changes.

Step 2: multiply the density values. Independence gives

$$L(f_1; \mathcal{D}_4) = \frac{1}{15} \cdot \frac{1}{15} \cdot \frac{1}{15} \cdot \frac{1}{15} = \frac{1}{50625},$$
$$L(f_2; \mathcal{D}_4) = \frac{1}{30} \cdot \frac{1}{30} \cdot \frac{1}{30} \cdot \frac{1}{30} = \frac{1}{810000}.$$

Thus the first model has the larger likelihood. Its score is

$$\frac{L(f_1; \mathcal{D}_4)}{L(f_2; \mathcal{D}_4)} = \left(\frac{30}{15}\right)^4 = 2^4 = 16$$

times the second model's score. Within this two-model class, the MLE selects $\widehat{f} = f_1$.

Step 3: understand the preference through probabilities. The first model concentrates its probability over a shorter interval, so it puts more probability near each observed value. For example, the interval $(3.9, 4.1)$ has width 0.2 minutes:

$$\Pr_{f_1}(3.9 < X < 4.1) = \frac{0.2}{15}, \qquad \Pr_{f_2}(3.9 < X < 4.1) = \frac{0.2}{30}.$$

Here $\Pr_{f_j}$ means probability calculated under candidate f_j. The first probability is twice the second. The same comparison holds for intervals of width 0.2 around 12, 7 and 3. By independence, the probability of all four observations falling in their respective intervals is $2^4 = 16$ times larger under f_1. This gives a probability interpretation of the likelihood ratio, even though the probability of any exact continuous observation is zero.

4.8.2 The empirical CDF and its likelihood interpretation

Definition 4.15: Empirical cumulative distribution function

Given observations $x_1, \ldots, x_n$, the **empirical CDF** is

$$\widehat{F}_n(t) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{\{x_i \le t\}}.$$

It is the fraction of observations at or below t. The corresponding empirical distribution assigns mass $1/n$ to each observation, counting repetitions.

To make the MLE interpretation precise, consider discrete distributions supported on the *distinct observed values* $z_1, \ldots, z_m$. Let n_j count how many observations equal z_j, and let w_j be the model's probability mass there. Then

$$w_j \ge 0, \qquad \sum_{j=1}^{m} w_j = 1, \qquad \sum_{j=1}^{m} n_j = n.$$

The exact-observation likelihood is

$$L(w_1, \ldots, w_m) = \prod_{j=1}^{m} w_j^{n_j}.$$

Maximizing it subject to the probability constraints gives

$$\widehat{w}_j = \frac{n_j}{n}.$$

Indeed, the log-likelihood is $\sum_j n_j \log w_j$. To impose the total-mass constraint while differentiating, form

$$\mathcal{L}(w, \lambda) = \sum_{j=1}^{m} n_j \log w_j - \lambda \left(\sum_{j=1}^{m} w_j - 1 \right).$$

The auxiliary variable λ is a **Lagrange multiplier**. Setting the derivatives in the weights to zero enforces the stationary condition alongside the constraint. Introducing a multiplier λ for $\sum_j w_j = 1$ gives $n_j/w_j = \lambda$; summing $w_j = n_j/\lambda$ yields $\lambda = n$. Concavity of the log-likelihood ensures that these weights maximize it. In fact, $d^2(n_j \log w_j)/dw_j^2 = -n_j/w_j^2 < 0$ for the distinct observed values, so this stationary point is the maximum on the probability simplex, the set of nonnegative weights summing to 1.

The resulting CDF is exactly

$$\sum_{j=1}^{m} \frac{n_j}{n} \mathbf{1}_{\{z_j \le t\}} = \widehat{F}_n(t).$$

This is the discrete nonparametric likelihood interpretation of the empirical distribution. It is not obtained by maximizing a product of CDF values, nor by freely maximizing over all continuous PDFs.

5

Stochastic simulations

5.1 A "what if?" experiment

A "what if?" question asks what would happen in a hypothetical situation. A simulation reproduces possible outcomes of an experiment on a computer. We can repeat the simulation and observe what happens across the different repetitions.

Example 28: One hundred coin tosses

Suppose we ask:

> If I tossed a coin 100 times, how many heads would I get?

A simulation produces a possible sequence of 100 tosses, and we count the heads in that sequence. We can repeat this experiment on the computer and compare the numbers of heads obtained.

5.2 From a real coin to a simulation

Imagine that we have a coin, but we do not know its probability of landing heads. For each toss, let $X = 1$ mean heads and $X = 0$ mean tails. Its distribution is described by

$$P(X = 1) = p, \qquad P(X = 0) = 1 - p,$$

where p is unknown. This is our **population model**: it describes the possible outcomes and their probabilities.

We decide to toss the coin 100 times. We assume that the tosses are independent and that each has the same probability p of heads. This is our **sampling scheme**. We call the tosses **IID**: independent and identically distributed. The outcome of one toss gives no information about the others, and all tosses follow the same distribution.

Together, the population model and the sampling scheme describe the **data generating process**: the mechanism that produces the sequence of outcomes we observe.

Suppose we observe 65 heads in our 100 tosses. We use the observed proportion of heads to estimate the unknown probability:

$$\widehat{p} = \frac{65}{100} = 0.65.$$

The hat means that $\widehat{p}$ is an estimate calculated from the data. It does not mean that the true p is exactly 0.65; another set of 100 tosses could give a different proportion. Using data to learn about an unknown model is called **inference**.

Now imagine that we choose the probability ourselves, for example $p = 0.5$, and ask the computer to produce 100 independent tosses with that probability. This is **simulation**: we specify the model and the sampling scheme, then generate data following those rules. We know the rules, but we do not fix the outcomes. One run could give 48 heads and another 53.

Note: Inference and simulation

Inference. We observe data and use them to learn about an unknown model.

$$\text{Observed data} \longrightarrow \text{Information about the model.}$$

Simulation. We choose a model and a sampling scheme, then generate data from them.

$$\text{Specified model and sampling scheme} \longrightarrow \text{Simulated data.}$$

For our coin, the simulation question is: what numbers of heads could we observe in 100 independent tosses if $p = 0.5$? Repeating the experiment on the computer lets us explore the variation in those numbers.

5.3 Where do the random numbers come from?

We want the computer to simulate our 100 coin tosses. A useful starting point is a sequence of numbers that behave like independent draws from a uniform distribution on $(0, 1)$. But where do these numbers come from?

5.3.1 Using a physical source

A **true random number generator** measures a physical source of randomness, such as electronic noise or a quantum phenomenon. The device processes those measurements to obtain numbers that a computer can use. The changing patterns of lava lamps are another example of a physical source.

We can distinguish two meanings of randomness:

- **Epistemic randomness:** we cannot predict an outcome because we lack information about the process.

- **Ontological randomness:** randomness is intrinsic to the physical phenomenon itself.

For simulation, the practical idea is to obtain numbers from a physical process. This requires a physical source and a device to read it. Collecting and processing measurements can take time. Repeating the experiment does not reproduce the same sequence of outcomes, although we can save the numbers already collected.

5.3.2 Using a formula

There is another approach: choose an initial value x_0 and a function f that calculates the next value from the current one. Apply the same function repeatedly. This is how a **pseudorandom number generator** works.

$$\underbrace{x_0}_{\text{initial value}} \longrightarrow \underbrace{f(x_0)}_{x_1} \longrightarrow \underbrace{f(f(x_0))}_{x_2} \longrightarrow \underbrace{f(f(f(x_0)))}_{x_3} \longrightarrow \cdots$$

First we calculate $x_1 = f(x_0)$. We then use x_1 as the input to the same function, obtaining $x_2 = f(x_1) = f(f(x_0))$, and continue in this way. In general, $x_{k+1} = f(x_k)$: each output becomes the input for the next step. Here f denotes the update rule.

The procedure is **deterministic**: once the starting value and the rule are fixed, the entire sequence is determined. A suitable generator is designed so that the sequence imitates the statistical behaviour of random draws from the distribution we need.

The starting value is called the **seed**. Such generators can be fast and require no additional physical device. To reproduce a run, we can keep the seed and the generator settings rather than store the whole sequence.

Note: Why the seed matters

With the same generator, settings and seed, we obtain the same sequence. This makes a simulation **reproducible**.

Imagine that a simulation gives an unexpected result. We can run it again with the same seed and inspect exactly the same simulated experiment. We can also compare two programs using the same inputs.

These pseudorandom numbers are the starting material from which we generate simulated coin tosses, die rolls and observations from other distributions.

5.4 What should a pseudorandom sequence imitate?

We can now make the goal of the generator more precise. We want its output to imitate independent draws from a uniform distribution on $(0, 1)$.

Definition 5.1: Pseudorandom number generator

In a simplified description, a *pseudorandom number generator* (PRNG) uses a deterministic function

$$f : (0, 1) \longrightarrow (0, 1)$$

and a starting value, or *seed*, $u_0 \in (0, 1)$ to generate

$$u_0, \quad u_1 = f(u_0), \quad u_2 = f(f(u_0)), \quad \ldots, \quad u_n = f^n(u_0).$$

The function is chosen so that the sequence imitates the statistical behaviour of an IID Uniform$(0, 1)$ sample. Here $f^n(u_0)$ means applying f repeatedly n times.

There are two parts to this goal:

- **Uniformity.** Intervals of the same length should receive approximately the same proportion of values. For example, in a sufficiently long sequence, about 10% of the values should fall in $(0.2, 0.3)$, and about 10% in $(0.7, 0.8)$.
- **Imitating independence.** The sequence should imitate draws in which observing one value gives no information about the others. We want to avoid systematic patterns between successive values.

For example, consider the repeating sequence

$$0.1,\ 0.2,\ 0.3,\ \ldots,\ 0.9,\ 0.1,\ 0.2,\ \ldots$$

It visits different parts of the interval, but its next value is easy to predict. Spreading values across the interval is only one part of what we require from a good generator.

Note: The apparent paradox

We know the seed and the rule, so the whole sequence is determined. Yet we design the generator so that its output statistically resembles independent uniform draws. Imitating an IID sample is the goal; for a fixed seed, the generated sequence remains deterministic.

5.5 The linear congruential generator

A concrete way to generate the next value is to multiply the current value, add a constant, and keep the remainder after division.

Definition 5.2: Linear congruential generator

A *linear congruential generator* (LCG) uses integer parameters α, β, and $M > 1$, and an integer seed $x_0 \in \{0, 1, \ldots, M - 1\}$. Its update rule is

$$x_{k+1} = f(x_k) = (\alpha + \beta x_k) \bmod M.$$

Here mod$M$ gives the remainder after division by M, so every state x_k belongs to $\{0, 1, \ldots, M-1\}$.

At each step, we multiply the current value by β, add α, and take the remainder. The resulting integer becomes the input for the next step.

Example 29: Following the sequence and finding its period

Choose small parameters so that we can follow the calculations:

$$\alpha = 1, \qquad \beta = 5, \qquad M = 16, \qquad x_0 = 2.$$

The first three updates are

$$\begin{aligned} x_1 &= (1 + 5 \cdot 2) \bmod 16 = 11, \\ x_2 &= (1 + 5 \cdot 11) \bmod 16 = 56 \bmod 16 = 8, \\ x_3 &= (1 + 5 \cdot 8) \bmod 16 = 41 \bmod 16 = 9. \end{aligned}$$

For example, $56 = 3 \cdot 16 + 8$, so its remainder is 8. Continuing with the same rule gives

$$\begin{aligned} 2 &\to 11 \to 8 \to 9 \to 14 \to 7 \to 4 \to 5 \\ &\to 10 \to 3 \to 0 \to 1 \to 6 \to 15 \to 12 \to 13 \to 2. \end{aligned}$$

Once 2 returns, the same rule produces 11 again, then 8, and so on: the sequence repeats. The **period** is the length of this repeating cycle.

Here we visit all 16 possible states before returning to the starting value. The period is therefore 16, the maximum possible for $M = 16$. In general, an LCG can have period M with appropriate choices of its parameters.

To rescale the integer states to numbers between 0 and 1, we can use

$$u_k = \frac{x_k}{M+1}.$$

For the first three generated states in the example, this gives

$$\frac{11}{17} \approx 0.647, \qquad \frac{8}{17} \approx 0.471, \qquad \frac{9}{17} \approx 0.529.$$

The output can be 0 when the remainder is 0, and is always less than 1. Rescaling changes the range of the values; the parameters determine how the sequence behaves statistically.

Note: Why the choice of parameters matters

We want the sequence to avoid repeating too soon. If we instead choose $\beta = 2$, keeping $\alpha = 1$, $M = 16$ and $x_0 = 2$, we get

$$2 \to 5 \to 11 \to 7 \to 15 \to 15 \to 15 \to \cdots$$

The generator gets stuck at 15, so the repeating cycle has period 1. Such a sequence would be unsuitable for simulating coin tosses.

A long period is one requirement for a useful generator. We also need its output to imitate uniformity and independence. Our small example illustrates the calculations and a full cycle; assessing statistical quality requires more than checking the period.

5.6 The LCG in R

We can translate the update rule into a short R function. The argument `n` specifies how many values we ask the function to generate. The arguments `a` and `b` correspond to α and β, `M` is the modulus, and `u0` supplies the initial seed.

To follow the calculations, we keep the small parameters from the previous example and generate three values:

```
lcg <- function(n = 3, a = 1, b = 5, M = 16, u0 = 2) {
  # Map an integer seed to a valid state between 0 and M - 1.
  u0 <- u0 %% M
  out <- rep(NA, n)

  for (i in 1:n) {
    u0 <- (a + b*u0) %% M
    out[i] <- u0 / (M + 1)
  }

  return(out)
}
```

5.6.1 Checking uniformity with a histogram

A histogram helps us see where numerical observations are concentrated. Imagine measuring many people's heights and marking each measurement as a point on a horizontal axis. With enough measurements, nearby points overlap, hiding how many values occur in each region. Stacking only exactly equal values does not solve the problem: two heights can be very close without being identical.

Instead, divide the range into adjacent intervals, called *bins*. Group together all measurements in each bin, then draw a bar above it. In a **count histogram**, the bar's height is the number of observations in that bin. A taller bar means that more observations fall in that interval. This makes concentrations, spread and the overall shape easier to see.[1]

The bin width matters. Very narrow bins may leave almost every observation on its own; very wide bins can merge different features. Trying several widths helps us see which patterns persist.

[1]This introduction follows Josh Starmer's StatQuest: Histograms, Clearly Explained.

For our LCG, the observations are the generated numbers. We want to check whether the LCG output resembles draws from Unif$(0, 1)$. We use a histogram to compare their distribution with the uniform distribution. To compare it with a PDF, we will change its scale: each bar's **area** will represent the fraction of values in its bin. We derive the corresponding height below.

In this example, we choose to generate 1000 values by setting `n = 1000`. We use seed 26 and parameters $a = 69069069$, $b = 12345$ and $M = 4096$:

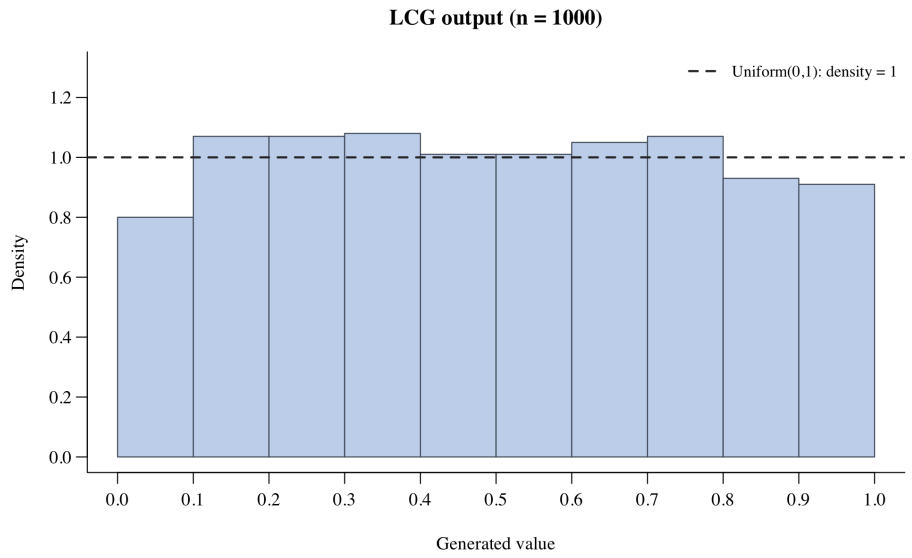
```
u <- lcg(n = 1000, a = 69069069, b = 12345,
         M = 2^12, u0 = 26)
```

The reasoning is as follows:

1. We ask the LCG to produce 1000 numbers between 0 and 1.
2. We want to see whether they are distributed approximately like a uniform distribution: in each interval of width 0.1, we expect about 100 numbers.
3. We build a **histogram of the numbers produced by the LCG** to see how many fall in each interval.
4. We compare it with what the uniform distribution predicts.

We divide $[0, 1]$ into ten bins of width 0.1. The uniform PDF is our theoretical reference. To compare the LCG output with it, we make each bar's **area** equal to the fraction of generated values in its bin. The calculation below explains how this determines the bar's height.

```
hist(u, breaks = seq(0, 1, 0.1), probability = TRUE)
abline(h = 1, lty = 2)
```



In this sequence, 80 values fall between 0 and 0.1, and 107 between 0.1 and 0.2, so the first bar is lower than the second.

With `probability = TRUE`, R scales the bars to represent an estimated PDF. To understand the calculation, consider the **first bin**, of width 0.1:

1. **Find the fraction of values in the bin.** There are 80 values in this bin out of 1000 in total, so
$$\frac{80}{1000} = 0.08.$$
This is 8% of the generated values, our estimate of the probability of falling in this bin.

2. **Give the bar this area.** Probabilities are areas under a PDF, so we want the rectangle's area to be 0.08. Since area equals width times height, we need
$$0.1 \times \text{height} = 0.08.$$

3. **Solve for the height.** Dividing the required area by the width gives
$$\text{height} = \frac{0.08}{0.1} = 0.8.$$

The first bar is therefore 0.8 high and has area $0.1 \times 0.8 = 0.08$, representing 8% of the values.

The same calculation applies to every bin. Let n be the total number of values, N_j the count in bin j, and h its width. The fraction in that bin is N_j/n; dividing this area by h gives

$$\text{height of bar } j = \frac{\text{fraction of values in bin } j}{\text{bin width}} = \frac{N_j/n}{h} = \frac{N_j}{nh}.$$

This is why we divide the count by *both* the sample size and the bin width. Every value belongs to one bin, so the fractions sum to 1 and the histogram's total area is 1.

The vertical axis shows the estimated PDF, that is, the bar heights. For a bin containing the expected 100 values, the height would be $100/(1000 \times 0.1) = 1$. The dashed line is the uniform PDF, 1 on $(0, 1)$. The bars are roughly close to it: a first visual check of uniformity. Their heights need not be identical in a finite sequence. A histogram does not check independence: rearranging the values leaves it unchanged.

5.7 Assessing the quality of a generator

We have seen how a generator produces numbers. We now ask whether those numbers behave well enough for our simulations.

Software uses generators more sophisticated than the LCG; the **Mersenne Twister** is one example. No pseudorandom generator produces a perfect IID sample: its sequence is determined by its initial state and update rule. Nevertheless, suitable generators can imitate independent uniform draws well enough for many practical purposes.

We assess three **properties** of the output, understood as criteria for judging the quality of the generator:

- **Property 0: Period.** The sequence should have a period long enough for the simulation. We examined this in the LCG example.
- **Property 1: Distribution of samples.** The values should imitate draws from a uniform distribution on $(0, 1)$. The histogram provides a first visual check.
- **Property 2: Independence of samples.** Groups of values should imitate independent draws, without systematic patterns between successive outputs.

Producing numbers between 0 and 1 is only the starting point; these properties concern the behaviour of the sequence as a whole.

5.7.1 Checking consecutive pairs

A histogram ignores the order of the values. To look for dependence, form the pairs

$$(u_1, u_2),\ (u_2, u_3), \ldots, (u_{n-1}, u_n)$$

and draw a scatter plot. Using the sequence `u` generated above, with at least two values:

```
n <- length(u)
plot(u[-n], u[-1], pch = ".",
     xlab = "Current value", ylab = "Next value")
```

Here `u[-n]` removes the last value, while `u[-1]` removes the first, so the vectors contain matching consecutive pairs. Lines, bands, or other systematic patterns reveal structure that the histogram may miss. A scatter without an obvious pattern is a visual diagnostic, not a proof of independence.

Note: For further reading

To prove when the LCG $x_{k+1} = (\alpha + \beta x_k) \bmod M$ has period M, we need to establish four points:

1. **First necessary condition.** Show that period M requires $\gcd(\alpha, M) = 1$.
2. **Second necessary condition.** Show that period M requires $\beta - 1$ to be divisible by every prime factor of M.
3. **Third necessary condition.** Show that, if 4 divides M, period M requires 4 to divide $\beta - 1$.
4. **Sufficiency.** Show that the three conditions together guarantee period M for every seed $x_0 \in \{0, 1, \ldots, M-1\}$.

Original reference: Hull and Dobell, *Random Number Generators* (1962).

5.8 The quantile transform method

We now have a way to generate numbers that imitate independent uniform draws. How can we use them to simulate a different distribution? The link is the quantile function.

Example 30: From uniform numbers to a biased coin

1. The goal. We want to simulate a coin whose result is 0 with probability 0.6 and 1 with probability 0.4. Let X denote the desired coin result:

$$P(X = 0) = 0.6, \qquad P(X = 1) = 0.4.$$

Each simulated toss must return one of these two values.

2. Start with a uniform number. The computer generates $U \sim \text{Unif}(0, 1)$. Split the unit interval into a part of length 0.6 and a part of length 0.4, and return

$$Y = \begin{cases} 0, & U \leq 0.6, \\ 1, & U > 0.6. \end{cases}$$

For example, $u = 0.25$ gives $y = 0$, whereas $u = 0.82$ gives $y = 1$. Since U is uniform, the first part is selected with probability 0.6 and the second with probability 0.4. The generated result Y therefore has the probabilities we wanted.

3. Recognize the quantile rule. The desired CDF satisfies

$$F(0) = P(X \leq 0) = 0.6, \qquad F(1) = P(X \leq 1) = 1.$$

The quantile $Q(p)$ returns the smallest value whose CDF reaches or exceeds the input probability level p. For $p = 0.82$, the value 0 does not qualify because $F(0) = 0.6 < 0.82$, while 1 does because $F(1) = 1 \geq 0.82$. Hence $Q(0.82) = 1$. For $p = 0.50$, the value 0 already qualifies because $F(0) = 0.6 \geq 0.50$, so $Q(0.50) = 0$.

4. The bridge. The input u is a probability level; the output $Q(u)$ is a possible value of the coin, 0 or 1. Using a randomly generated uniform number as that input gives $Y = Q(U)$. This turns uniform draws into draws from the desired distribution.

Previously, we chose a probability level p and found the corresponding threshold $Q(p)$. Now we draw that level at random: we generate U uniformly between 0 and 1 and return $Q(U)$. This is the **quantile transform method**, also called the **inverse transform method**.

Proposition 5.1: Quantile transform

Let F be any CDF and let

$$Q(p) = \inf\{x \in \mathbb{R} : F(x) \geq p\}, \qquad 0 < p < 1,$$

be its quantile function. If $U \sim \text{Unif}(0, 1)$ and $Y = Q(U)$, then Y has CDF F:

$$F_Y(y) = P(Y \leq y) = F(y), \qquad y \in \mathbb{R}.$$

Proof in the continuous, strictly increasing case. In this case $Q = F^{-1}$. Since F is increasing, $F^{-1}(U) \leq y$ is equivalent to $U \leq F(y)$. Therefore,

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P\big(F^{-1}(U) \leq y\big) \\ &= P\big(U \leq F(y)\big) \\ &= F(y). \end{aligned}$$

The last step uses the standard uniform distribution's CDF: $P(U \leq t) = t$ for $0 \leq t \leq 1$. Here the threshold is $t = F(y)$, which lies in $[0, 1]$ because F is a CDF. Thus the generated variable has exactly the CDF we wanted. The result also holds for CDFs with jumps or flat parts, using the general quantile definition; the proof above covers the invertible case.

To generate a sample, draw independent $U_1, \ldots, U_n$ and set $Y_i = Q(U_i)$ for each i. Applying the same function separately to independent draws preserves independence, and every Y_i has CDF F.

5.8.1 A continuous example

Example 31: Generating values with density $3x^2$

Suppose we want to simulate a variable X with PDF

$$f_X(x) = \begin{cases} 3x^2, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the CDF. For $0 \le x < 1$,

$$F_X(x) = \int_0^x 3t^2\,dt = [t^3]_0^x = x^3.$$

The complete CDF is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x^3, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

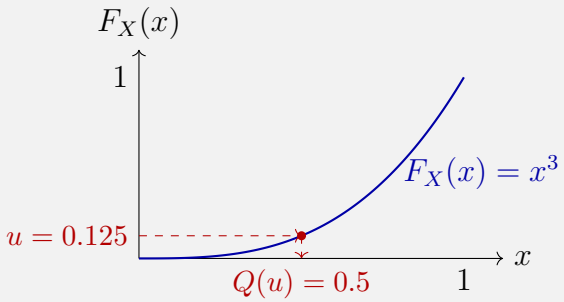
Find the quantile. For $0 < p < 1$, solve $x^3 = p$:

$$Q(p) = p^{1/3}.$$

Generate a value. Draw $U \sim \text{Unif}(0, 1)$ and return

$$\boxed{Y = U^{1/3}.}$$

For example, the uniform value $u = 0.125$ becomes $y = 0.125^{1/3} = 0.5$.



Geometrically, start at the randomly chosen height u, move right to the CDF, and read the corresponding value on the horizontal axis.

As a check, for $0 \le y \le 1$,

$$P(Y \le y) = P(U^{1/3} \le y) = P(U \le y^3) = y^3 = F_X(y).$$

5.8.2 A discrete example

Example 32: Generating a Bernoulli variable

Recall the Bernoulli distribution introduced in Section 3.9.3. Suppose we want $X \sim \text{Ber}(0.4)$, so

$$P(X = 0) = 0.6, \qquad P(X = 1) = 0.4.$$

Its CDF is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 0.6, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

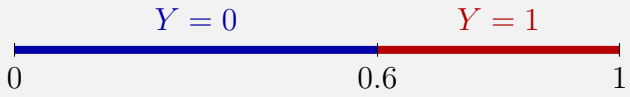
Find the quantile. This CDF has jumps. We use the smallest threshold where the CDF reaches or exceeds the chosen probability level:

$$Q(p) = \begin{cases} 0, & 0 < p \le 0.6, \\ 1, & 0.6 < p < 1. \end{cases}$$

Generate a value. Draw $U \sim \text{Unif}(0, 1)$ and return

$$Y = Q(U) = \begin{cases} 0, & U \le 0.6, \\ 1, & U > 0.6. \end{cases}$$

The interval for output 0 has length 0.6; the interval for output 1 has length 0.4.



For example, $u = 0.25$ gives $y = 0$, whereas $u = 0.8$ gives $y = 1$. At $u = 0.6$, the general quantile definition gives $y = 0$.

Check the probabilities. Since U is uniform,

$$P(Y = 0) = P(U \le 0.6) = 0.6, \qquad P(Y = 1) = P(U > 0.6) = 1 - 0.6 = 0.4.$$

Thus Y has the same distribution as X. Equivalently, $Y = \mathbf{1}_{(0.6,1)}(U)$: the indicator is 1 when U falls above 0.6, and 0 otherwise.

5.8.3 Population generation in R

The continuous example can be implemented directly. Use the uniform sequence `u` generated earlier and apply its quantile function to every value:

```
fx <- function(x) ifelse(x >= 0 & x <= 1, 3*x^2, 0)
Fx <- function(x) ifelse(x < 0, 0, ifelse(x < 1, x^3, 1))
Qx <- function(p) p^(1/3)

y <- Qx(u)
hist(y, breaks = 20, probability = TRUE, ylim = c(0, 3.2))
curve(fx(x), from = 0, to = 1, add = TRUE, lwd = 2)
```

The three functions represent the target PDF, CDF, and quantile, respectively. The quantile input must lie in $[0, 1]$; its values at the endpoints are assigned by continuous extension. The histogram of `y` is compared with $3x^2$, not with the horizontal uniform density.

R also provides a built-in uniform generator:

```
set.seed(26)
u <- runif(1000)
y <- Qx(u)
```

`set.seed` makes the generated sequence reproducible; `runif` produces pseudorandom uniform values. The transformation step is the same whichever suitable uniform generator supplies the input.

5.9 Transformations of random variables

5.9.1 What are we trying to compute?

Suppose we know the distribution of a random variable X and apply a deterministic function r to its value. This defines a new random variable

$$Y = r(X).$$

The function r tells us how to transform each value. We want to find the **distribution of the resulting values**: the CDF F_Y and, when it exists, the PDF f_Y.

In the quantile transform method, we chose $r = Q$ to obtain a desired distribution from uniform draws. Here we study what distribution a given transformation produces.

Example 33: Squaring a uniform random variable

Let $X \sim \text{Unif}(-1, 1)$ and define $Y = X^2$. The PDF of X, from Section 3.9.1, is

$$f_X(x) = \frac{1}{2}\mathbf{1}_{[-1,1]}(x).$$

For example, $X = -0.5$ gives $Y = 0.25$, and $X = 0.5$ gives the same result. Squaring loses the sign of the input.

Since $-1 \leq X \leq 1$, we have $0 \leq Y \leq 1$. Every value in $[0, 1]$ can be obtained by squaring a value in $[-1, 1]$, so the output range is $[0, 1]$. Knowing this range alone does not tell us how probability is distributed within it.

5.9.2 Find probabilities first, then the CDF

First calculate one interval probability, then let the threshold vary to obtain the full CDF.

5.9.2.1 Start with a concrete threshold

For the example above, consider the probability that the *transformed* value is at most 0.25:

$$P(Y \leq 0.25).$$

To compute it, find which *input* values produce such an output. Because $Y = X^2$,

$$Y \leq 0.25 \quad \Longleftrightarrow \quad X^2 \leq 0.25 \quad \Longleftrightarrow \quad -0.5 \leq X \leq 0.5.$$

Both negative and positive input values must be included. The selected interval has length 1, while the support $[-1, 1]$ has length 2. Uniformity of X therefore gives

$$P(Y \leq 0.25) = P(-0.5 \leq X \leq 0.5) = \frac{1}{2}.$$

Thus half the probability of Y lies in the first quarter of its range. If Y were uniform on $[0, 1]$, that probability would be 0.25. **The transformed variable is not uniform.**

5.9.2.2 Replace the threshold by a general value

The CDF asks the same question for every threshold y:

$$F_Y(y) = P(Y \leq y).$$

For $0 \leq y \leq 1$, solving $X^2 \leq y$ gives $-\sqrt{y} \leq X \leq \sqrt{y}$. Hence

$$\begin{aligned} F_Y(y) &= P(-\sqrt{y} \leq X \leq \sqrt{y}) \\ &= \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2}\, dx = \frac{\sqrt{y} - (-\sqrt{y})}{2} = \sqrt{y}. \end{aligned}$$

Including thresholds outside the support gives the complete CDF:

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ \sqrt{y}, & 0 \leq y \leq 1, \\ 1, & y > 1. \end{cases}$$

As a check, $F_Y(0.25) = \sqrt{0.25} = 0.5$, agreeing with the concrete calculation. This is a cumulative probability, not a PDF value.

5.9.3 From the CDF of Y to the PDF of Y

For a variable with a PDF, the CDF accumulates area under the PDF:

$$F_Y(y) = \int_{-\infty}^{y} f_Y(t)\, dt.$$

Differentiation recovers the PDF from this accumulated area. For our example, on the interior of the output range,

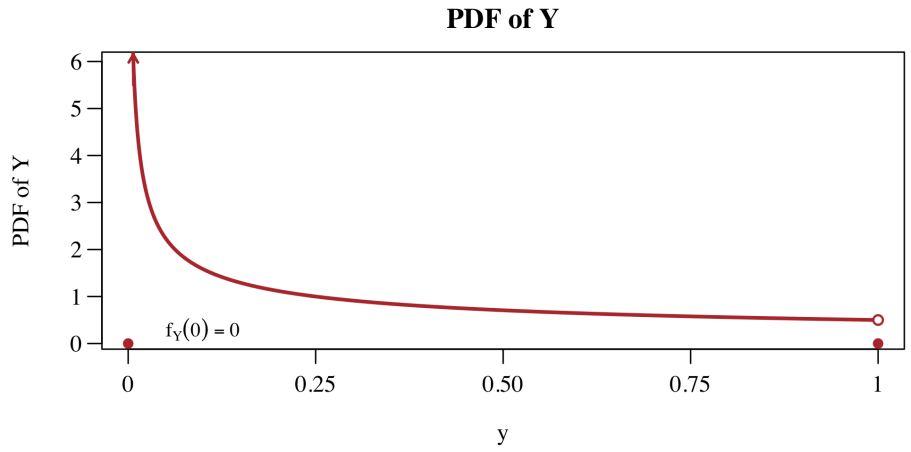
$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy}\sqrt{y} = \frac{1}{2\sqrt{y}}, \qquad 0 < y < 1.$$

A complete version of the PDF is

$$f_Y(y) = \begin{cases} \dfrac{1}{2\sqrt{y}}, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

For a continuous random variable with a PDF, probabilities are areas under the PDF. A single point has zero width, so changing the value of the PDF only at $y = 0$ or $y = 1$ does not change any area or probability.

The formula $1/(2\sqrt{y})$ is undefined at $y = 0$, because it would require division by zero. We choose $f_Y(0) = 0$ to complete the definition. This is the value at the exact point 0. For positive values of y closer and closer to 0, the formula gives larger and larger PDF values, without any upper bound.



5.9.3.1 Why is the PDF high near zero?

Squaring brings values with absolute value below 1 closer to zero, and both signs contribute to the same output values. The calculation $P(0 \leq Y \leq 0.25) = 0.5$ shows the resulting concentration near zero.

The high PDF is not a positive probability at the point 0: $P(Y = 0) = P(X = 0) = 0$.

Interval probabilities are areas under the PDF. For $0 \leq a < b \leq 1$,

$$P(a \leq Y \leq b) = F_Y(b) - F_Y(a) = \sqrt{b} - \sqrt{a}.$$

For example,

$$P(0.25 \leq Y \leq 1) = 1 - 0.5 = 0.5.$$

Compare the two intervals:

- $[0, 0.25]$ has length 0.25 and contains probability 0.5.
- $[0.25, 1]$ has length 0.75 and also contains probability 0.5.

The same amount of probability is therefore collected in a shorter interval near zero. This shows that the probability of Y is more concentrated near zero.

The PDF grows without bound near zero. Could its total area therefore be infinite? As we know, the total area should be 1, as required for a PDF. The height of the curve alone does not determine its area: we must calculate the integral.

To handle the unbounded PDF near zero, start at a positive number ε:

$$\int_{\varepsilon}^{1} \frac{1}{2\sqrt{y}}\, dy = \left[\sqrt{y}\right]_{\varepsilon}^{1} = 1 - \sqrt{\varepsilon}.$$

Then let ε approach zero from the right, including more and more of the area near zero:

$$\int_{0}^{1} \frac{1}{2\sqrt{y}}\, dy = \lim_{\varepsilon \downarrow 0} (1 - \sqrt{\varepsilon}) = 1.$$

Thus the PDF grows without bound near zero, but its total area is finite and equals 1.

5.9.4 The general method: transform the event

We now generalize the procedure used in the example $Y = X^2$ so that we can apply it to any transformation $Y = r(X)$. We start from the known PDF of X and want to find the CDF of Y and, when it exists, its PDF.

The procedure follows three steps. Suppose X has a PDF f_X, support S_X, and $Y = r(X)$.

1. **Find the input values selected by a threshold.** For a fixed y, solve the inequality $r(x) \leq y$ on the support of X. Write its solution set as

$$A_y = \{x \in S_X : r(x) \leq y\}.$$

   The subscript records that this set changes when the threshold y changes.

2. **Compute the CDF using the input distribution.** The events $\{Y \leq y\}$ and $\{X \in A_y\}$ are the same, so

$$F_Y(y) = P(r(X) \leq y) = P(X \in A_y) = \int_{A_y} f_X(x)\, dx.$$

   We integrate the known PDF of X over the selected input values to obtain a probability about Y.

3. **Obtain a PDF when one exists.** If the CDF can be written as an integral of a PDF, differentiate it to recover $f_Y(y) = F_Y'(y)$ wherever the derivative exists.

Note

This method does not require r to be invertible.

Note

Even if X has a PDF, its transformation Y may not have one.

For example, let $X \sim \text{Unif}(-1, 1)$ and choose $r(x) = 0$. Every input value gives $Y = 0$, so $P(Y = 0) = 1$.

If Y had a PDF, we would instead have

$$P(Y = 0) = P(0 \leq Y \leq 0) = \int_0^0 f_Y(t)\, dt = 0.$$

An integral with identical lower and upper limits is zero. This contradicts $P(Y = 0) = 1$, so Y has no PDF. Its PMF assigns $p_Y(0) = 1$. The event method still gives its CDF: $F_Y(y) = 0$ for $y < 0$ and $F_Y(y) = 1$ for $y \geq 0$.

5.9.5 When the transformation has an inverse

There is another way to carry out the calculation in the previous section when r is strictly monotone and has an inverse.

We can use the known CDF of X directly to find the CDF of $Y = r(X)$, without explicitly calculating the integral over A_y.

The inverse lets us rewrite the same condition $r(X) \leq y$ as a condition on X that its CDF can evaluate.

Suppose r is strictly monotone on the input support: it is either strictly increasing or strictly decreasing. Each value in the output range then corresponds to exactly one input value. The inverse $s = r^{-1}$ returns that input: $s(y)$ is the unique value satisfying $r(s(y)) = y$. For thresholds y inside the output range, applying the inverse lets us express the event using X directly.

If r is increasing, applying the inverse preserves the inequality:

$$F_Y(y) = P(r(X) \leq y) = P(X \leq s(y)) = F_X(s(y)).$$

If r is decreasing, applying the inverse reverses it:

$$F_Y(y) = P(X \geq s(y)) = 1 - F_X(s(y)),$$

where we used continuity of X, so $P(X = s(y)) = 0$.

Under the usual smoothness assumptions (for example, r is continuously differentiable with nonzero derivative on the interior of the support), we obtain the PDF by differentiating the CDF. The chain rule gives $f_X(s(y))s'(y)$ in the increasing case and $-f_X(s(y))s'(y)$ in the decreasing case. Since $s'(y) > 0$ in the first case and $s'(y) < 0$ in the second, both expressions can be written as

$$f_Y(y) = f_X(s(y))\,|s'(y)|.$$

The factor $|s'(y)|$ accounts for the change in interval lengths.

Example 34

This single-inverse formula cannot be applied directly to x^2 on the full interval $[-1, 1]$. Indeed, $r'(x) = 2x$, so $r'(0) = 0$: the assumption of a nonzero derivative on the interior of the support is not satisfied. Moreover, $r(-0.5) = r(0.5) = 0.25$, so r has no single inverse on this interval.

5.9.5.1 The connection with the quantile transform

For the square example, $F_Y(y) = \sqrt{y}$ on $[0, 1]$ is strictly increasing. Solving $p = \sqrt{y}$ for y gives

$$Q_Y(p) = p^2, \qquad 0 < p < 1.$$

For example, $Q_Y(0.5) = 0.25$, the threshold containing half the probability below it.

Thus the quantile transform offers another way to generate this distribution: draw $U \sim \text{Unif}(0,1)$ and return $Q_Y(U) = U^2$. Both X^2 with $X \sim \text{Unif}(-1,1)$ and U^2 with $U \sim \text{Unif}(0,1)$ have the CDF $\sqrt{y}$ on $[0,1]$.

5.9.6 Check the transformation in R

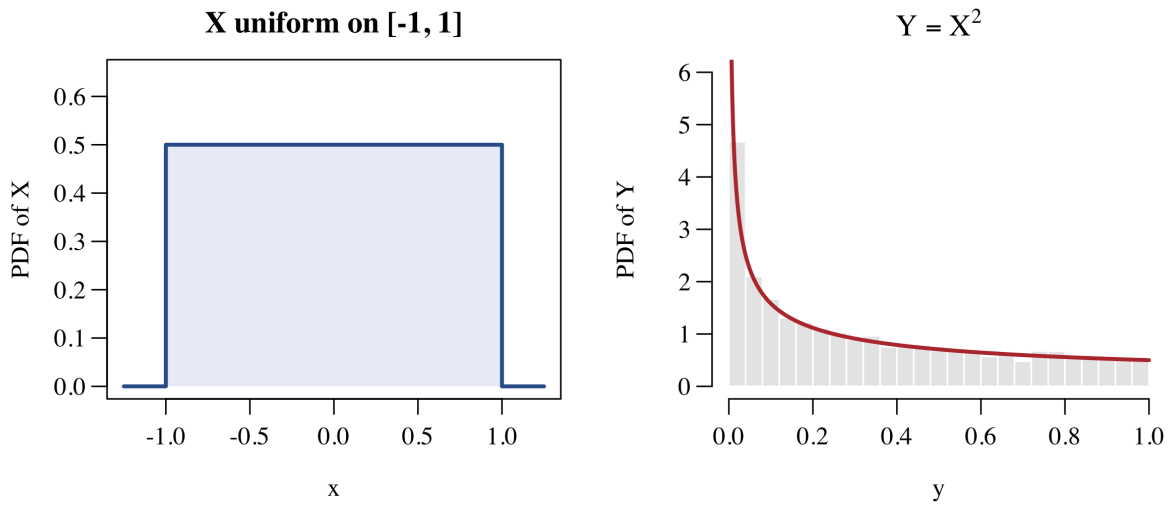
To check our calculation, we generate values of X, square each one to obtain values of Y, and compare the output histogram with the PDF we derived:

```
set.seed(8)
x <- runif(5000, min = -1, max = 1)
y <- x^2

hist(y, breaks = seq(0, 1, length.out = 26),
     probability = TRUE, xlab = "y",
     ylab = "PDF of Y", main = "Y = X^2")
curve(1 / (2 * sqrt(x)), from = 0.001, to = 1,
      add = TRUE, col = "red", lwd = 2)
```

`runif` produces 5000 pseudorandom uniform values in $[-1, 1]$. The operation `x^2` squares each element of the vector. With `probability = TRUE`, the histogram estimates the PDF of Y, so its bar areas estimate interval probabilities.

Inside `curve`, `x` is the plotting argument: it represents a possible output value y in the PDF formula. The curve is drawn only inside the output range and starts above zero to avoid division by zero.



The left panel shows the PDF of X.

The right panel compares the transformed sample with $1/(2\sqrt{y})$.

The curve continues upwards beyond the displayed range near zero; the finite height of the first histogram bar estimates the average PDF value over that bin.

6

Exploratory data analysis

We now use observed samples to learn a population distribution. The empirical CDF and likelihood were introduced after IID sampling. The simulation chapter then supplied a concrete histogram check. Here we develop density estimation, ways to measure its error, and the role of the class of candidate models.

6.1 From generator checks to statistical learning

The histogram check for a generator leads to a more general question: how can we measure the difference between a target density and an estimate obtained from data?

6.1.1 A target density and a histogram estimate

For the standard uniform distribution, the target is known:

$$f_U(u) = \mathbf{1}_{[0,1]}(u).$$

Let $u_1, \ldots, u_n$ be the available values. A density histogram with equal bin width h gives an estimate $\widehat{f}_{n,h}$. If bin B_j contains N_j observations, its height is

$$\widehat{f}_{n,h}(u) = \frac{N_j}{nh}, \qquad u \in B_j.$$

Use disjoint bins covering the range, assigning a boundary point to just one bin. Each bin's area is N_j/n, so the histogram has total area 1.

The notation records two choices: n is the sample size and h is the bin width. The histogram is an algorithm that turns the data into a density estimate.

6.1.2 Distances between densities

Consider the following ways to compare densities f and g:

$$D_1(f,g) = \int_{\mathbb{R}} |f(u) - g(u)|\, du,$$
$$D_2(f,g) = \left(\int_{\mathbb{R}} |f(u) - g(u)|^2\, du \right)^{1/2},$$
$$D_\infty(f,g) = \operatorname*{ess\,sup}_{u \in \mathbb{R}} |f(u) - g(u)|.$$

These are the L^1, L^2, and L^∞ distances, when finite. The first accumulates absolute discrepancies; the second squares them before integration; the third looks at the largest discrepancy, ignoring changes on sets of length zero.

More generally, for $1 \le q < \infty$,

$$D_q(f,g) = \left(\int_{\mathbb{R}} |f(u) - g(u)|^q\, du \right)^{1/q}.$$

Different distances emphasize different features of the error.

6.1.3 Loss for one dataset and average risk

An average over random outcomes is called an **expectation**. For a nonnegative random variable Z, it is a probability-weighted average: $\mathbb{E}[Z] = \sum_z z\, p_Z(z)$ in a discrete model, or $\mathbb{E}[Z] = \int_{\mathbb{R}} z f_Z(z)\, dz$ when Z has a PDF. The value may be $+\infty$. We apply this idea to the nonnegative error produced by a random dataset.

Once the data are fixed, the discrepancy

$$\ell_n(u_1, \ldots, u_n) = D\big(f_U, \widehat{f}_{n,h}\big)$$

is the **loss** for that dataset. A different dataset may produce a different histogram and a different loss.

The blackboard contrasted two possible goals:

- **Worst-case loss:** control the largest loss over the possible datasets. In schematic form,
$$\sup_{\underline{u} \in [0,1]^n} \ell_n(\underline{u}) \longrightarrow 0.$$
- **Average loss:** average over datasets generated by a specified sampling model. This average is the **risk**:
$$R_n(f_U; h) = \mathbb{E}_{f_U}\Big[D\big(f_U, \widehat{f}_{n,h}\big)\Big].$$

For example, under ideal independent uniform sampling, the expectation averages over the random samples $(U_1, \ldots, U_n)$. Risk tending to zero is an average guarantee; it does not require small loss for every possible dataset. More explicitly, for IID observations from a density f and an estimation rule $\widehat{f}_{n,h}$, this average is

$$R_n(f; h) = \int_{\mathbb{R}^n} D\big(f, \widehat{f}_{n,h}(\,\cdot\,; x_1, \ldots, x_n)\big) \prod_{i=1}^{n} f(x_i)\, dx_1 \cdots dx_n.$$

The dot is the argument of the estimated density; the values $x_1, \ldots, x_n$ specify the dataset used to construct it. The product is the joint PDF supplied by the IID model.

The worst-case requirement is much stronger. For an ordinary histogram it need not hold: datasets concentrated in one bin remain possible even when the target distribution is uniform.

Note: A criterion is not an LCG convergence theorem

These formulas express ways to judge an estimation procedure. They do not prove that every LCG converges to a uniform density. With a fixed seed, an LCG produces a deterministic sequence; an average risk requires a stated source of randomness, such as a random seed or a sampling model. Results for ideal IID data do not automatically apply to a deterministic generator.

6.1.4 Unknown ground truth and a class of models

In the simulation check, f_U is known. In a learning problem, the true population density f is typically unknown, and the algorithm must estimate it from data.

For data generated from f, define

$$R_n(f; h) = \mathbb{E}_f\Big[D\big(f, \widehat{f}_{n,h}\big)\Big].$$

The subscript f records the distribution used to average over datasets. The phrase "all ground truths that make sense" refers to a specified class of possible densities $\mathcal{F}$.

There are two distinct consistency requirements:

- **For a fixed density:** $R_n(f; h_n) \to 0$ for that population density f.
- **Uniformly over a class:**
$$\sup_{f \in \mathcal{F}} R_n(f; h_n) \longrightarrow 0.$$

In the second expression, we first average over datasets for each f, then take the worst risk over the class of population densities. This supremum is over *models*, whereas the earlier worst-case loss was over *datasets*.

The class $\mathcal{F}$ and the estimation procedure matter. The formula is a goal to establish under suitable assumptions, not a guarantee for every class of densities. Writing h_n allows the bin width to change with sample size; a fixed bin width may retain approximation error for a general density.

6.1.5 Qualitative and quantitative questions

The final board remarks distinguish:

- **Qualitative:** does the risk tend to zero? This is a consistency question.
- **Quantitative:** how fast does it tend to zero as the sample size grows? This asks for a convergence rate.

A rate may be expressed by a bound such as

$$\sup_{f \in \mathcal{F}} R_n(f; h_n) \le C r_n, \qquad r_n \longrightarrow 0,$$

where C does not depend on n. Such a bound must be proved for the chosen model class, loss, and algorithm; these formulas do not establish a specific rate for the LCG.

6.2 Density estimation

6.2.1 Which distributions can we choose from?

Before choosing a model, we need to decide which distributions we will consider. Each candidate gives a possible explanation of how the observations were generated. A *class of models* is simply the set of candidates we allow ourselves to choose from.

Definition 6.1: Class of models

In this density-estimation problem, a **class of models** $\mathcal{F}$ is a set of candidate PDFs. Each $f \in \mathcal{F}$ represents one possible population distribution.

We compare these candidates using the maximum likelihood rule introduced in Section 4.8.1. The chosen class $\mathcal{F}$ specifies which densities can compete in that comparison.

With this idea in mind, we now ask why we cannot simply allow maximum likelihood to choose among all possible densities f.

6.2.2 Why not consider every possible PDF?

Now we can explain the difficulty. If we allow **every possible PDF**, we can build a curve with very narrow peaks centred exactly on our observed values. Making the peaks narrower lets us make them taller while keeping the total area equal to 1.

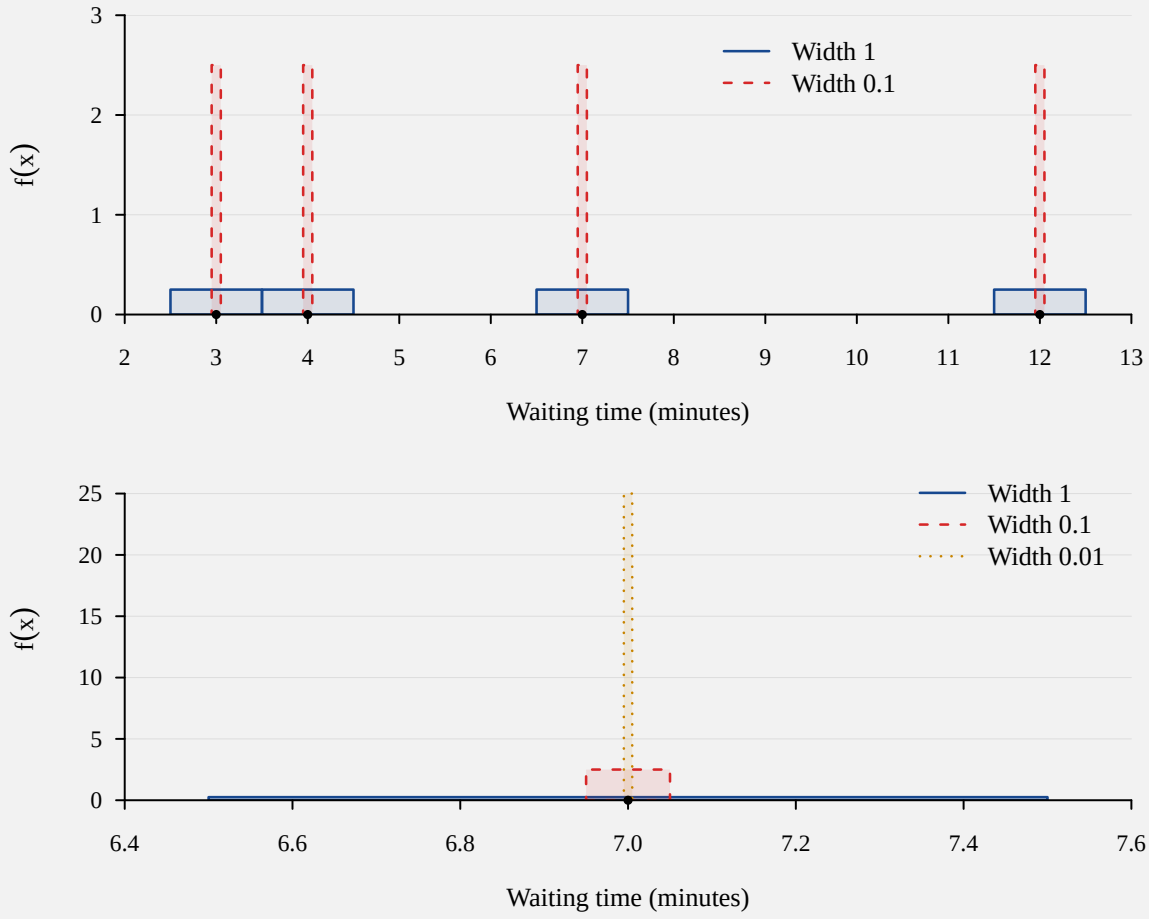
As the peaks become taller, the values $f(x_i)$ increase, so their product increases too. We can keep increasing the likelihood without reaching a largest finite value. The instruction "choose the PDF with the highest likelihood" therefore fails to select a PDF in this unrestricted search.

Example 35

Use our four observed waiting times, measured in minutes:

$$\mathcal{D}_4 = (4, 12, 7, 3).$$

We construct a candidate PDF specifically around these data: four small rectangles, each centred on one observed value. Each rectangle has area $1/4$, so their total area is 1. Outside the rectangles, the density is zero.



Step 1: keep the area fixed while narrowing the rectangles. The upper panel shows all four data points. The lower panel zooms in on the rectangle centred at 7. Each colour represents a different candidate PDF. We can make the rectangles narrower while keeping each area equal to 1/4, provided we increase their height:

$$\text{height} = \frac{\text{area}}{\text{width}}.$$

For the widths shown, the rectangles have disjoint interiors. The numerical comparison is:

Width of each rectangle	Height	Area of each rectangle
1	0.25	$1 \cdot 0.25 = 1/4$
0.1	2.5	$0.1 \cdot 2.5 = 1/4$
0.01	25	$0.01 \cdot 25 = 1/4$

In every row, the four areas still sum to 1: every candidate is a valid PDF.

Step 2: evaluate the density at the observed values. Each observation lies at the centre of its own rectangle, so the PDF value there equals that rectangle's height.

If the common width is h, with $0 < h \leq 1$, denote the resulting density by f_h. At each of the four observations,

$$f_h(4) = f_h(12) = f_h(7) = f_h(3) = \frac{1}{4h}.$$

Thus narrowing every rectangle by a factor of 10 multiplies each observed density value by 10.

Step 3: multiply the four density values. Under the IID model, the likelihood is

$$L(f_h; \mathcal{D}_4) = f_h(4) f_h(12) f_h(7) f_h(3) = \left(\frac{1}{4h}\right)^4.$$

When the rectangles become ten times narrower, this product increases by

$$10 \cdot 10 \cdot 10 \cdot 10 = 10^4 = 10\,000.$$

We can keep narrowing the rectangles and increasing their heights, always preserving total area 1. The likelihood grows without bound as $h \to 0^+$, so this unrestricted search has no largest finite likelihood. This is why we need to control which densities are allowed to compete.

Note: How can we stop MLE from increasing the likelihood by building ever-narrower peaks around the data?

To make the search meaningful, we need to control which curves can win. One approach is to choose a restricted family of PDFs before fitting the model. Another is to change the comparison score so that overly irregular curves receive a penalty. The class of models and the fitting rule together define the learning problem we are trying to solve.

6.2.3 How does a penalty enter the comparison?

First, take the logarithm of the likelihood. This turns the product into a sum* without changing which positive likelihood is largest:

$$\log L(f; \mathcal{D}_n) = \sum_{i=1}^{n} \log f(x_i).$$

Working with this sum has two practical advantages:

1. **More reliable numerical calculations.** Suppose we have 1000 observations and each contributes a factor of 0.1 to the likelihood. Their product is $0.1^{1000} = 10^{-1000}$, so small that ordinary computer arithmetic may round it to zero. Instead, we can calculate

$$\log L = 1000 \log(0.1) \approx -2302.6,$$

*For $a, b > 0$, $\log(ab) = \log a + \log b$.

using the natural logarithm. We add the log of each factor directly, without first computing the tiny product.

2. **Easier differentiation.** To find a maximum analytically, we often differentiate with respect to the model parameter. For a sum, we differentiate each term separately and add the results. This is usually simpler than applying the product rule to a likelihood with hundreds or thousands of factors.

We now subtract a penalty from the log-likelihood. The result is called the **penalized log-likelihood**.

Definition 6.2: Penalized log-likelihood

For fixed observed data $\mathcal{D}_n = (x_1, \ldots, x_n)$ from an IID sample, consider a candidate PDF f. Let $J(f) \geq 0$ be its penalty, and let $\lambda > 0$ be the weight given to that penalty. The **penalized log-likelihood** is

$$\log L(f; \mathcal{D}_n) - \lambda J(f) = \sum_{i=1}^{n} \log f(x_i) - \lambda J(f).$$

If two densities have the same log-likelihood, we prefer the one with the smaller penalty.

Example 36

Return to Example 35, with the same four waiting times $4, 12, 7, 3$. We keep the four rectangles, each with area $1/4$, but now add a cost for making them tall.

Step 1: choose the penalty. For this example, let the penalty be the height of the tallest rectangle. All four rectangles have the same height, which we call H, so $J(f) = H$. Choose $\lambda = 1$: we subtract the height itself from the log-likelihood.

Step 2: calculate the new result. Each observation lies inside a rectangle of height H. The likelihood is therefore H^4, and its logarithm is $4 \log H$. After subtracting the penalty, we obtain

$$\underbrace{4 \log H}_{\text{log-likelihood}} - \underbrace{H}_{\text{penalty}}.$$

Compare the three widths already shown in Example 35. The values below use natural logarithms and are rounded to two decimal places.

Width	Height H	Log-likelihood	Penalized log-likelihood
1	0.25	-5.55	-5.80
0.1	2.5	3.67	**1.17**
0.01	25	12.88	-12.12

Among these three candidates, we now choose the rectangles of width 0.1.

Step 3: see why narrower is no longer always better. Going from width 0.1 to width 0.01 increases the log-likelihood by about 9.21, but increases the penalty by $25 - 2.5 = 22.5$. The extra height costs more than it gains.

This also stops the problem when we allow any common width $0 < h \leq 1$. The height is $H = 1/(4h)$, so shrinking h towards zero makes H grow without limit. However, the penalty H eventually grows faster than $4 \log H$:

$$4 \log H - H \longrightarrow -\infty \qquad \text{as } H \longrightarrow +\infty.$$

Within this family of rectangular densities, making the peaks ever taller can no longer keep improving the result.

A

Crash course exercises

A.1 Day 01

Exercise 1: Computer virus

A virus can enter a computer through e-mail or the internet. The probability of entry through e-mail is 0.30, through the internet 0.40, and through both routes 0.15. What is the probability that the virus does not enter at all?

Solution.

Let

$$E = \{\text{the virus enters through e-mail}\}, \qquad I = \{\text{the virus enters through the internet}\}.$$

Step 1: Define the target event. Let $N = E^c \cap I^c$ be the event that the virus does not enter. We seek $P(N)$.

Step 2: Apply De Morgan's law.

$$P(N) = P\big((E \cup I)^c\big).$$

Step 3: Use the complement rule.

$$P\big((E \cup I)^c\big) = 1 - P(E \cup I).$$

Step 4: Calculate the union by inclusion–exclusion.

$$P(E \cup I) = P(E) + P(I) - P(E \cap I) = 0.30 + 0.40 - 0.15 = 0.55.$$

Step 5: Take the complement.

$$\boxed{P(N) = 1 - 0.55 = 0.45 = 45\%.}$$

Exercise 2: Two children

A family has two children, and at least one is a boy. What is the probability that both are boys?

Solution.

Assume each child is independently a boy or a girl with probability 1/2. Let

$$B_1 = \{\text{the first child is a boy}\}, \qquad B_2 = \{\text{the second child is a boy}\}.$$

Step 1: Define the events in the question. Let

$$A = B_1 \cap B_2 = \{\text{both children are boys}\},$$
$$C = B_1 \cup B_2 = \{\text{at least one child is a boy}\}.$$

We seek $P(A \mid C)$.

Step 2: List the ordered outcomes. With B for boy and G for girl, the equally likely possibilities are

$$\Omega = \{BB, BG, GB, GG\}.$$

Step 3: Use the information given. The event C excludes GG, leaving

$$C = \{BB, BG, GB\}.$$

Step 4: Count the favorable outcomes. We have $A = \{BB\}$, so one of the three outcomes in C is favorable:

$$\boxed{P(A \mid C) = \frac{1}{3}.}$$

Check using conditional probability.

Since $A \subseteq C$, we have $A \cap C = A$. Independence and the complement rule give

$$P(A) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}, \qquad P(C) = 1 - P(\{GG\}) = 1 - \frac{1}{4} = \frac{3}{4}.$$

Therefore

$$P(A \mid C) = \frac{P(A \cap C)}{P(C)} = \frac{1/4}{3/4} = \frac{1}{3}.$$

If the information had instead been *the first child is a boy*, only BB and BG would remain, giving 1/2. The way the information is supplied matters.

A.2 Day 02

Exercise 3: Rolling two dice

Two fair six-sided dice are rolled. Describe the sample space and find the probability that their sum is seven.

Solution.

Step 1: Write the sample space. The result is an ordered pair (i, j), one value for each die:

$$\Omega = \{(i, j) : i, j \in \{1, \dots, 6\}\}, \qquad |\Omega| = 6 \cdot 6 = 36.$$

In particular, $(1, 6)$ and $(6, 1)$ are different outcomes.

Step 2: Define the target event. Let

$$S = \{(i, j) \in \Omega : i + j = 7\} = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}.$$

We seek $P(S)$, and $|S| = 6$.

Step 3: Apply the equally likely outcomes rule.

$$\boxed{P(S) = \frac{|S|}{|\Omega|} = \frac{6}{36} = \frac{1}{6}.}$$

Exercise 4: Five-card poker hands

Assume that all five-card hands from a standard 52-card deck are equally likely. Find the probabilities of a flush (five cards of one suit), four of a kind, and exactly one pair.

Solution.

Because order does not matter, all three probabilities have the same denominator:

$$N = \binom{52}{5} = 2\,598\,960.$$

Flush.

Step 1: Choose the suit: 4 possibilities.

Step 2: Choose five cards of that suit: $\binom{13}{5}$ possibilities.

Step 3: Count and divide:

$$\boxed{P(\text{flush}) = \frac{4\binom{13}{5}}{N} = \frac{5\,148}{2\,598\,960} \approx 0.001981.}$$

Under the exercise's definition, this includes straight flushes.

Four of a kind.

Step 1: Choose the rank: 13 possibilities. All four cards of that rank are then fixed.

Step 2: Choose the fifth card: any of the remaining 48.

Step 3: Count and divide:

$$\boxed{P(\text{four of a kind}) = \frac{13 \cdot 48}{N} = \frac{624}{2\,598\,960} \approx 0.0002401.}$$

Exactly one pair.

Step 1: Choose the pair's rank: 13 possibilities.

Step 2: Choose its two suits: $\binom{4}{2}$ possibilities.

Step 3: Choose the other three ranks: $\binom{12}{3}$ ways. They must be distinct, avoiding another pair or a triple.

Step 4: Choose their suits: 4^3 possibilities.

Step 5: Count and divide:

$$\boxed{P(\text{exactly one pair}) = \frac{13\binom{4}{2}\binom{12}{3}4^3}{N} = \frac{1\,098\,240}{2\,598\,960} \approx 0.42257.}$$

A.3 Day 04 (24 sept 26)

Exercise 5: Rome and Florence

The probability that a tourist has visited Rome is 26%, the probability that they have visited Florence is 18%, and the probability that they have visited at least one of the two cities is 38%.

Question A. What is the probability that the tourist has visited both Rome and Florence?

Question B. What is the probability that the tourist has visited Rome but not Florence?

Notation.

Let

$$R = \{\text{the tourist has visited Rome}\}, \qquad F = \{\text{the tourist has visited Florence}\}.$$

The given probabilities are

$$P(R) = 0.26, \qquad P(F) = 0.18, \qquad P(R \cup F) = 0.38.$$

Solution to A.

Step 1: Define the target event. Let

$$A = R \cap F = \{\text{the tourist has visited both cities}\}.$$

We seek $P(A)$.

Step 2: Apply inclusion–exclusion and substitute.

$$\begin{aligned} P(A) &= P(R) + P(F) - P(R \cup F) \\ &= 0.26 + 0.18 - 0.38 \\ &= \boxed{0.06 = 6\%}. \end{aligned}$$

Solution to B.

Step 1: Define the target event. Let

$$B = R \cap F^c = \{\text{the tourist has visited Rome but not Florence}\}.$$

We seek $P(B)$.

Step 2: Subtract the overlap from Rome.

$$\begin{aligned} P(B) &= P(R) - P(A) \\ &= 0.26 - 0.06 \\ &= \boxed{0.20 = 20\%}. \end{aligned}$$

Exercise 6: An uncommon disease

You are diagnosed with an uncommon disease. You know that there is only a 1% chance of getting it. Let

$$D = \{\text{you have the disease}\}, \qquad T = \{\text{the test is positive}\}.$$

It is known that the test is imperfect, since

$$P(T \mid D) = 0.98, \qquad P(T^c \mid D^c) = 0.95.$$

Question A. Given that you test positive, what is the probability that you have the disease?

Question B. You take the test again, independently of the first test conditional on whether you have the disease, and it is positive again. Given both positive results, what is the probability that you have the disease?

Solution to A.

We seek $P(D \mid T)$. The prior probabilities are $P(D) = 0.01$ and $P(D^c) = 0.99$.

Step 1: Find the false-positive rate. By the complement rule,

$$P(T \mid D^c) = 1 - P(T^c \mid D^c) = 1 - 0.95 = 0.05.$$

Step 2: Find the probability of a positive result. By the law of total probability,

$$\begin{aligned} P(T) &= P(T \mid D)P(D) + P(T \mid D^c)P(D^c) \\ &= 0.98 \cdot 0.01 + 0.05 \cdot 0.99 \\ &= 0.0593. \end{aligned}$$

Step 3: Apply Bayes' theorem.

$$\boxed{P(D \mid T) = \frac{P(T \mid D)P(D)}{P(T)} = \frac{0.0098}{0.0593} \approx 16.53\%.}$$

The positive test raises the probability of disease from 1% to about 16.53% under the stated assumptions.

Solution to B.

Let

$$T_1 = \{\text{the first test is positive}\},$$
$$T_2 = \{\text{the second test is positive}\}.$$

We seek $P(D \mid T_1 \cap T_2)$.

Step 1: Find the likelihood of two positive results. Conditional independence gives

$$P(T_1 \cap T_2 \mid D) = 0.98^2 = 0.9604, \qquad P(T_1 \cap T_2 \mid D^c) = 0.05^2 = 0.0025.$$

Step 2: Find the probability of two positive results. By the law of total probability,

$$\begin{aligned} P(T_1 \cap T_2) &= P(T_1 \cap T_2 \mid D)P(D) + P(T_1 \cap T_2 \mid D^c)P(D^c) \\ &= 0.9604 \cdot 0.01 + 0.0025 \cdot 0.99 \\ &= 0.012079. \end{aligned}$$

Step 3: Apply Bayes' theorem.

$$\boxed{P(D \mid T_1 \cap T_2) = \frac{0.9604 \cdot 0.01}{0.012079} \approx 79.51\%.}$$

The second positive result raises the probability from about 16.53% to about 79.51% under the conditional independence assumption.

B

Random variable exercises

B.1 A density given up to a constant

Exercise 7: Normalize, condition and transform

Let X have a PDF specified up to a positive constant:

$$f_X(x) \propto x\,\mathbf{1}_{[0,2]}(x).$$

(1) Find its PDF, CDF and quantile $Q_X(1/4)$.

(2) Find the conditional density given $A = \{X > 1\}$.

(3) Find the distribution of $Y = \sqrt{X}$ under the original distribution of X.

The symbol $\propto$ means that the PDF is proportional to the given kernel: a positive multiplicative constant is still unknown. The indicator is 1 on $[0, 2]$ and 0 elsewhere, restricting the density to that interval. With our convention that the support contains the points of positive density, its support is $(0, 2]$.

Part (1): normalize the kernel.

The kernel is $g(x) = x\mathbf{1}_{[0,2]}(x)$. Its integral is

$$c = \int_0^2 x\,dx = \left[\frac{x^2}{2}\right]_0^2 = 2.$$

Dividing by this total gives the PDF

$$\boxed{f_X(x) = \frac{x}{2}\mathbf{1}_{[0,2]}(x).}$$

The density increases with x: intervals of the same small width near the right end carry more probability than those near zero.

Part (1): find the CDF and a quantile.

For $0 \le x \le 2$, the event $X \le x$ retains the part of the support from 0 to x. Consequently,

$$F_X(x) = \int_0^x \frac{t}{2}\, dt = \frac{x^2}{4}.$$

Including thresholds outside the support gives the complete CDF:

$$\boxed{F_X(x) = \begin{cases} 0, & x < 0, \\ x^2/4, & 0 \le x \le 2, \\ 1, & x > 2. \end{cases}}$$

We seek the quantile at probability $1/4$: a threshold below which one quarter of the probability lies. Since this CDF is strictly increasing on $[0, 2]$, solve

$$F_X(x) = \frac{1}{4} \iff \frac{x^2}{4} = \frac{1}{4} \implies \boxed{Q_X(1/4) = 1.}$$

More generally, $Q_X(u) = 2\sqrt{u}$ for $0 < u < 1$. A continuous CDF need not always have an ordinary inverse; here strict increase on $[0, 2]$ justifies this inversion.

Note: What happens where the density is zero?

If the PDF is zero throughout an interval, the CDF stays constant there: integrating over that interval adds no probability. For example, a density supported on $[0, 1] \cup [3, 4]$ has a flat CDF between 1 and 3. It can still be a continuous distribution, but its CDF is not strictly increasing everywhere. This explains why the generalized quantile definition is useful.

Part (2): condition on $A = \{X > 1\}$.

This is conditioning on an event of positive probability, rather than on an exact value of a continuous variable. We keep only the part of the density on $1 < x \le 2$, then normalize it within that interval. First,

$$P(A) = \int_1^2 \frac{x}{2}\, dx = \frac{3}{4}.$$

For any measurable set B, the conditional probability is $P(X \in B \mid A) = P(X \in B, A)/P(A)$. Thus its density is

$$\boxed{f_{X|A}(x) = \frac{f_X(x)\mathbf{1}_{(1,2]}(x)}{P(A)} = \frac{2x}{3}\mathbf{1}_{(1,2]}(x).}$$

It integrates to 1 on the retained interval. The handwritten calculation reaches the same result by first normalizing the restricted kernel $x\mathbf{1}_{(1,2]}(x)$, whose integral is $3/2$.

Part (3): transform the original variable.

We now return to the original distribution of X. Since $0 \le X \le 2$, the output satisfies $0 \le Y \le \sqrt{2}$. For y in that range, taking the square root is increasing, so

$$F_Y(y) = P(\sqrt{X} \le y) = P(X \le y^2) = F_X(y^2) = \frac{y^4}{4}.$$

Differentiation gives

$$\boxed{f_Y(y) = y^3 \mathbf{1}_{[0,\sqrt{2}]}(y).}$$

The complete CDF is zero for $y < 0$ and one for $y > \sqrt{2}$. We can also apply the inverse transformation formula from Section 5.9.5. The inverse of $y = \sqrt{x}$ is $x = y^2$, whose derivative is $2y$:

$$f_Y(y) = f_X(y^2) \left| \frac{d}{dy} y^2 \right| = \frac{y^2}{2}\, 2y = y^3, \qquad 0 < y < \sqrt{2}.$$

Both approaches produce the same density and the same support.

B.2 Two throws of a die

Exercise 4.1 — Two throws of a die

Source: Dekking et al., A Modern Introduction to Probability and Statistics, Section 4.6, p. 51.

Throw a fair die twice, independently. Let Z count the number of times a 6 appears. For an outcome $\omega = (\omega_1, \omega_2)$, define

$$S(\omega_1, \omega_2) = \omega_1 + \omega_2, \qquad M(\omega_1, \omega_2) = \max\{\omega_1, \omega_2\}.$$

(a) Describe the distribution of Z by giving either its PMF p_Z or its CDF F_Z. Identify the distribution family and its parameters.

(b) List the outcomes in $\{M = 2, Z = 0\}$, $\{S = 5, Z = 1\}$, and $\{S = 8, Z = 1\}$, and calculate their probabilities.

(c) Determine whether $\{M = 2\}$ and $\{Z = 0\}$ are independent.

Part (a): the PMF of Z

Since Z counts the number of sixes in two throws, Z takes values in $\{0, 1, 2\}$. Formally, it is the random variable

$$Z : \{1, \dots, 6\}^2 \longrightarrow \mathbb{R}, \qquad Z(\omega_1, \omega_2) = \mathbf{1}_{\{\omega_1 = 6\}} + \mathbf{1}_{\{\omega_2 = 6\}}.$$

Here each indicator is 1 if the corresponding throw is 6, and 0 otherwise.

To calculate the PMF, we proceed in two steps.

1. **First, show that Z is discrete.** Choose the finite set $\mathcal{X} = \{0, 1, 2\}$. Every outcome produces one of these values, so

   $$P(Z \in \mathcal{X}) = P(\Omega) = 1.$$

   Therefore, Z is discrete.

2. **Next, calculate its PMF.** By definition, $p_Z(x) = P(Z = x)$ for $x \in \mathbb{R}$. There are 36 equally likely ordered outcomes.
   If $x = 0$, neither throw is 6: each throw has 5 possible values.

   $$P(Z = 0) = \frac{5 \cdot 5}{36} = \frac{25}{36}.$$

   If $x = 1$, exactly one throw is 6: it can be the first or the second.

   $$P(Z = 1) = \frac{1 \cdot 5 + 5 \cdot 1}{36} = \frac{10}{36}.$$

   If $x = 2$, both throws are 6: the only outcome is $(6, 6)$.

   $$P(Z = 2) = \frac{1}{36}.$$

Hence its probability mass function (PMF) is:

$$p_Z(x) = \begin{cases} 25/36, & x = 0, \\ 10/36, & x = 1, \\ 1/36, & x = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Part (a): calculating the CDF

Alternatively, we can use the cumulative distribution function,

$$F_Z(x) := P(Z \leq x).$$

Here $\{Z \leq x\} = \{\omega \in \Omega : Z(\omega) \leq x\}$: we collect the outcomes for which the number of sixes is at most x.

Since Z takes only the values $0, 1, 2$, the included values change only when x reaches one of these numbers. We therefore distinguish $x < 0$, $0 \leq x < 1$, $1 \leq x < 2$, and $x \geq 2$.

For $x < 0$, no possible value of Z is included, so

$$F_Z(x) = P(Z \leq x) = P(\varnothing) = 0.$$

For $0 \leq x < 1$, only $Z = 0$ is included:

$$F_Z(x) = P(Z \leq x) = P(Z = 0) = \frac{25}{36}.$$

In particular, $F_Z(0) = 25/36$.

For $1 \leq x < 2$, the included values are $Z = 0$ and $Z = 1$:

$$\{Z \leq x\} = \{Z = 0\} \cup \{Z = 1\}.$$

These events are disjoint: Z cannot equal both 0 and 1 in the same outcome. By additivity of probability for disjoint events,

$$F_Z(x) = P(Z \leq x) = P(Z = 0) + P(Z = 1) = \frac{25}{36} + \frac{10}{36} = \frac{35}{36}.$$

In particular, $F_Z(1) = 35/36$.

For $x \geq 2$, all three possible values are included:

$$\{Z \leq x\} = \{Z = 0\} \cup \{Z = 1\} \cup \{Z = 2\} = \Omega.$$

These three events are pairwise disjoint. Therefore,

$$\begin{aligned} F_Z(x) = P(Z \leq x) &= P(Z = 0) + P(Z = 1) + P(Z = 2) \\ &= \frac{25}{36} + \frac{10}{36} + \frac{1}{36} = \frac{36}{36} = 1, \qquad x \geq 2. \end{aligned}$$

Putting the cases together, its CDF is:

$$F_Z(x) = \begin{cases} 0, & x < 0, \\ 25/36, & 0 \leq x < 1, \\ 35/36, & 1 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$

Part (b): outcomes and probabilities

Both conditions in each event must hold for the same ordered pair. Each of the 36 ordered outcomes has probability 1/36.

1. $\{M = 2, Z = 0\}$. If the maximum is 2, both throws are at most 2 and at least one is 2. All these outcomes also give $Z = 0$:

   $$\{M = 2, Z = 0\} = \{(1, 2), (2, 1), (2, 2)\}.$$

   Therefore,

   $$P(M = 2, Z = 0) = \frac{3}{36} = \frac{1}{12}.$$

2. $\{S = 5, Z = 1\}$. Exactly one throw must be 6. The other is at least 1, so the sum is at least 7 and cannot be 5:

   $$\{S = 5, Z = 1\} = \varnothing, \qquad P(S = 5, Z = 1) = 0.$$

3. $\{S = 8, Z = 1\}$. One throw is 6 and the other must be 2:

   $$\{S = 8, Z = 1\} = \{(6, 2), (2, 6)\}.$$

   Therefore,

   $$P(S = 8, Z = 1) = \frac{2}{36} = \frac{1}{18}.$$

Part (c): independence

Let $A = \{M = 2\}$ and $B = \{Z = 0\}$. By the definition of independence, we must check whether $P(A \cap B) = P(A)P(B)$.

If $M = 2$, neither throw can be 6, so $Z = 0$. Thus $A \subseteq B$ and $A \cap B = A$. Using the counts above,

$$P(A \cap B) = P(A) = \frac{3}{36} = \frac{1}{12}.$$

On the other hand,

$$P(A)P(B) = \frac{3}{36} \cdot \frac{25}{36} = \frac{25}{432} \neq \frac{1}{12}.$$

Therefore, $\{M = 2\}$ and $\{Z = 0\}$ are **not independent**. Equivalently, knowing that $M = 2$ guarantees that $Z = 0$:

$$P(Z = 0 \mid M = 2) = 1 \neq \frac{25}{36} = P(Z = 0).$$